CP RENEWAL VALIDATION REPORT FOR UZUNDERE I 63.0 MW HYDROELECTRIC POWER PLANT PROJECT, TURKEY

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Summary:

Re Carbon Ltd. has performed the VCS crediting period (CP) renewal validation of the UZUNDERE I 63.0 MW HYDROELECTRIC POWER PLANT PROJECT, TURKEY (UZUNDERE HEPP) in Turkey between 10/04/2022 and 06/01/2023.

Uzundere HEPP involves installation and operation of licensed hydro power project and the total capacity of the project is 63.068 MWm / 62.152 MWe ((31.534 MWm / 31.076 MWe) X 2). During the second crediting period which is 10 years, the project is expected to reduce 76,997 tons of CO_2 annually.

The purpose of this validation activity is to determine whether the project complies with the applicable rules and regulations and to see whether the project still can be eligible to be a VCS project activity.

The scope of the validation is the independent and objective review of the VCS-PD against the relevant criteria. The main focus of the validation team is to identify whether the project is still eligible to be a VCS project activity, whether to check the baseline scenario is still valid, whether to check the emission reduction calculations are handled correctly and to whether to be able to continue with the generation of VCUs through the emission reduction.

The validation was performed on the basis of VCS criteria, Host Party criteria as well as the criteria of Clean Development Mechanism methodologies and tools. An online (remote) audit was performed on 18/04/2022. During the validation 04 CARs and 07 CLs were raised. All of which have resolved before the end of the validation.

The review of the Project Description (VCS-PD) and the subsequent follow-up interviews have provided Re Carbon Ltd. with sufficient evidence to determine the fulfillment of all stated criteria. In our opinion, the project meets all relevant VCS requirements. Therefore, Re Carbon Ltd. recommends the renewal of CP of project by the VCS Program.

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1 INTRODUCTION

1.1 Objective

Re Carbon Ltd. has been appointed by “KARADENİZ HES ELEKTRİK A.Ş.” to perform the crediting period (CP) renewal validation of the “UZUNDERE I 63.0 MW HYDROELECTRIC POWER PLANT PROJECT, TURKEY” with the service agreement dated 18/02/2022. The objective of this validation activity is to have an independent third party for the assessment of the project design, and to ensure a thorough assessment of the proposed project activity against the applicable VCS and CDM requirements. In particular;

- the project's baseline is assessed against “ACM0002 version 20.0: “Consolidated baseline methodology for grid-connected electricity generation from renewable sources”
- the project’s monitoring plan is assessed against “ACM0002 version 20.0: “Consolidated baseline methodology for grid-connected electricity generation from renewable sources”
- the projects compliance with, the requirements of Article 12 of the Kyoto Protocol, the CDM Modalities and Procedures as agreed in the Marrakech Accords under decision 3/CMP.1, the annexes to this decision, subsequent decisions and guidance made by COP/MOP & CDM Executive Board and other relevant rules, including the Host Country legislation and sustainability criteria
- CDM Validation and Verification Standard version 3.0
- CDM Project Standard version 3.0
- CDM Project Cycle Procedure version 3.0
- VCS Version 4.0
- VCS Program Guide version 4.0

CP renewal validation is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of voluntary emission reductions (VERs).

1.2 Scope and Criteria

The scope of the validation is the independent and objective review of the VCS Project Description (PD). The PD is reviewed against the relevant criteria (see section 1.1) and decisions by the VCS Organization, including the approved baseline and monitoring methodology. The validation was based on the guidance given in the CDM Validation and Verification Standard for project activities, version 3.0, CDM Project Standard for project activities, version 3.0 and CDM Project Cycle Procedure for project activities, version 3.0 and VCS Version 4.0.

The validation team has employed a risk based approach to assess the completeness and accuracy of the claims and conservativeness of the assumptions in the PD. The main focus of the validation team is to identify whether the project is still eligible to be a VCS project activity, whether to check the baseline scenario is still valid, whether to check the emission reduction calculations are handled correctly and to whether to be able to continue with the generation of VCUs through the emission reduction. The validation is not meant to provide any consulting towards the project participants. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the project further implementation.

The only purpose of crediting period renewal validation is its usage during the crediting period renewal process as part of the VCS project cycle. Therefore, Re Carbon Ltd. can't be held liable by any party for decisions made or not made based on the validation opinion, which will go beyond that purpose.

1.3 Level of Assurance

Level of assurance of the validation report is defined as reasonable based on the handled validation process through the document review, online (remote) audit and the interviews.

1.4 Summary Description of the Project

KARADENİZ HES ELEKTRİK A.Ş. has installed a hydro power plant (Uzundere HEPP) in Black Sea Region of Turkey, within the province of Rize, Turkey. Uzundere HEPP involves installation and operation of a licensed project. Uzundere HEPP project entails the construction and operation of a 63.068 MWm / 62.152 MWe hydroelectric power plant. The project consists of hydro power plant with two turbines of 31.534 MWm / 31.076 MWe capacity.

The project is in operation and started to produce electricity on 27/05/2010. The project fits in sectoral scope 1: Energy Industries -Renewable Energy category.

The electricity production of the proposed project is expected to be 156,205 MWh/year during the second crediting period. Since the project activity is a renewable energy project, it reduces greenhouse gases (GHGs) emissions that would have otherwise occurred in the absence of the project activity by avoiding electricity generation from fossil fuel sources. In the baseline scenario, the electricity delivered to the grid by the project would have otherwise been generated by the operation of grid-connected power plants. Since Turkey's grid mainly consists of thermal power plants, this would have resulted in GHG emissions. However, in the project scenario, the project will continue to generate electricity from hydro power and will result in emission reductions in parallel with its electricity generation figures. The average annual emission reductions of the proposed project for the crediting period are estimated to be 76,997 tonnes of CO_2e (tCO_2e). The emission reduction calculations were validated by the VVB via a detailed review of the baseline calculation Excel sheets.

2 VALIDATION PROCESS

Re Carbon Ltd. has performed the CP renewal validation of the "UZUNDERE I 63.0 MW HYDROELECTRIC POWER PLANT PROJECT, TURKEY" between 10/04/2022 and 06/01/2023. The validation was performed on the basis of VCS criteria, UNFCCC criteria for the CDM and Host Party criteria, as well as criteria given to provide for consistent project operations, monitoring and reporting as detailed in below.

2.1 Method and Criteria

The scope of the CP renewal validation is the independent and objective review of the VCS PD and the baseline calculations together with the supporting documents that are listed in section 2.2 of this report. This review is done according to rules and regulations set forth in the applied methodology, associated tools, program standards, guidelines and procedures.

During the review, the project's compliance with the host country's relevant laws and regulations are also assessed and deemed as appropriate. As the project is a hydro power plant, no sampling was deemed necessary and the validation process includes the whole project.

2.2 Document Review

The basis for the validation activity is the PD version 01, dated 14/04/2022 which was submitted to the validation team on the same day. This PD was revised several times due to the raised CARs and CLs, version 4 dated 06/01/2023 being the final version. The PD was assessed against;

- the project's baseline is assessed against "ACM0002-Version 20.0: "Consolidated baseline methodology for grid-connected electricity generation from renewable sources" and
- Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period, Version 3.0.1
- the projects compliance with, the requirements of Article 12 of the Kyoto Protocol, the CDM Modalities and Procedures as agreed in the Marrakech Accords under decision 3/CMP.1, the annexes to this decision, subsequent decisions and guidance made by COP/MOP & CDM Executive Board and other relevant rules, including the Host Country legislation and sustainability criteria
- CDM Validation and Verification Standard version 3.0
- CDM Project Standard version 3.0
- CDM Project Cycle Procedure version 3.0
- VCS Version 4.0
- VCS Program Guide version 4.1
- the Host Country criteria

The list of the documents which were reviewed during the CP renewal validation process is given in the Table 2-1 below:

Table 2-1: List of documents reviewed

Document Number	Document Name	Version	Date (dd/mm/yyyy)
D01	Revised PD	01	14/04/2022
D02	Revised PD	02	12/05/2022
D03	Registered PD	04	22/11/2010
D04	Validation Report	02	17/03/2011
D05	Initial Verification Report	1.1	03/12/2021
D06	ER Calculation Excel Spreadsheet	01	14/04/2022
D07	ER Calculation Excel Spreadsheet	02	12/05/2021
D08	ACM0002: Grid-connected electricity generation from renewable sources	20	29/11/2019
D09	CDM Validation and Verification Standard version	3.0	09/09/2021

Document Number	Document Name	Version	Date (dd/mm/yyyy)
D10	CDM Project Standard version	3.0	09/09/2021
D11	CDM Project Cycle Procedure version	3.0	09/09/2021
D12	VCS Standard	4.0	22/04/2021
D13	VCS Program Guide	4.0	19/09/2019
D14	CP Renewal Validation Service Agreement	-	18/02/2022
D15	National Emission factor of Turkey	-	01/09/2020
D16	Meters Test Reports and Photographic Evidences of Electricity Meters	-	15/10/2020
D17	Social Security Records of PP Site Staff	-	-
D18	Training Records	-	-
D19	Letter by Village Heads (Mukhtar) (About the Contact Details of PP Relevant Staff In case of Any Complaint)	-	26/02/2021
D20	Hazardous Waste Disposal Records	-	-
D21	Generation License (Last Amendment)	-	31/08/2006
D22	EIA Positive Decision	-	01/04/2010
D23	Provisional Acceptance Protocol	-	14/08/2009
D24	Area of Project	-	-
D25	Fish passage photos	-	-
D26	Minimum water monitoring and official declaration	-	17/02/2022
D27	Waste management evidences	-	2012-2017
D28	Double counting declarations	-	26/02/2021
D29	Revised PD	03	18/05/2022
D30	ER Calculation Excel Spreadsheet	03	18/05/2022
D31	Revised PD	04	06/01/2023

2.3 Interviews

During the CP renewal validation period, follow-up interviews were realized by the validation team to further analyze the correctness and accurateness of the information provided.

The list of people who were interviewed during the CP renewal validation process is given in the Table 2-2 below:

Table 2-2: List of persons interviewed

Reference Number	Means of Interview[1]	Full Name	Title	Organization
I01	SV	Yusuf Sukas	Muhtar	Incesu Village
I02	SV	Hasan Yüksel	Resident	Incesu Village
I03	SV	Ummuhan Yüksel	Resident-Female	Incesu Village
I04	SV	Celal Kork	Resident	Cataldere Village
I05	SV	Zeynep Kork	Resident-Female	Cataldere Village
I06	SV	Osman Köse	Plant Manager	KARADENİZ HES ELEKTRİK A.Ş.

2.4 Site Inspections

As a part of the CP renewal validation activities, the physical site visit hadn't been performed due to COVID-19 outbreak but the following had been implemented and remote audit was handled on 18/04/2022 and through Zoom program, details of which can be seen in the Table 2-3 below:

Table 2-3: Remote Audit details

Date	18/04/2022	
Location	Remote audit through Zoom program	
Participant	**Company Name**	**Role in the Organization / Role in the Site Visit**
Yusuf Sukas	Incesu Village	Muhtar
Hasan Yüksel	Incesu Village	Resident
Ummuhan Yüksel	Incesu Village	Resident-Female
Celal Kork	Cataldere Village	Resident
Zeynep Kork	Cataldere Village	Resident-Female
Osman Köse	KARADENİZ HES ELEKTRİK A.Ş.	Plant Manager
Points Verified	**Source of Information**	
To check the project development and operation	Document review and remote audit	
To interview with the local stakeholders about the project and its impacts	Remote audit and interviews with the local stakeholders from İncesu and Çataldere Villages	
To confirm rightness of project description, as per PD including project components and location	Document review, remote audit and interviews with the local stakeholders from İncesu and Çataldere Villages	

The project owner representatives and local stakeholders had been interviewed as in above during the remote audit using Zoom program and some photographic evidences like electricity meters, waste storage areas etc. have been taken along with the document review process to

[1] SV: Online Site visit; T: Telephone; E: E-mail

achieve the reasonable level of assurance during the CP renewal validation as detailed in other sections of the report.

2.5 Resolution of Findings

The crediting period renewal validation of the mentioned VCS project activity includes the following phases:

- Assessment whether the project design of the proposed VCS project activity meets the relevant VCS requirements, via a desk review of the PD between 10/04/2022 and 06/01/2023.
- Assessment of the stakeholders' comments and how these comments are implemented in the PD
- Assessment whether the applied methodology ACM0002-Version.20.0: "Consolidated baseline methodology for grid-connected electricity generation from renewable sources", has been applied correctly, including the baseline selection and monitoring plan.
- An remote audit was conducted on 18/04/2022 to assess the implementation process of the project activity and to confirm stakeholders' comments.
- Assessment of data and calculation of greenhouse gas emission reductions.
- Issuance of the validation report
- Independent technical review (ITR)
- Approval of the validation report and request of CP renewal

The Validation Protocol is used for the assessment of each requirement during the execution of validation activities and is given in Annex-1 of this validation report.

The Validation Protocol consists of two tables:

- Table 1 (Project Description and VCS validation requirements) and
- Table 2 (Resolution of Corrective Action, Forward Action and Clarification Requests)

The usage description of Table-1 in Validation Protocol is explained in Table 2-4 below:

Table 2-4: Explanation about Table-1 in Validation Protocol

Question	Reference	MoV*	Findings, comments, references and document sources	Draft & Final Conclusion
The requirements related with the VCS project	Gives reference to the legislation or documents	Explains how conformance with question is investigated. Examples of	Is used to elebarote and discuss the question and/or conformance to	Either acceptable based on the evidence provided (OK), non-compliance with the

Question	Reference	MoV*	Findings, comments, references and document sources	Draft & Final Conclusion
description and validation	where the relevant requirement is found	means of validation are Document Review (DR), Interview (I) and Not Applicable (NA)	the question by giving related references and document sources based on which the finding is issued or evidence is checked	requirement (CAR), further clarification (CL) due to insufficient, unclear or not transparent information, forward action request (FAR) that needs to be solved during the first verification

The usage description of Table-2 in Validation Protocol is explained in Table 2-5 below:

Table 2-5: Explanation about Table-2 in Validation Protocol

Draft Report Clarifications, Forward Action and Corrective Action Requests by Validation Team	Ref. to Questions in Table-1	Summary of Project Participants' Response	Validation Team Conclusion
The all CL, FAR and CARs determined during the draft validation report should be listed here	Gives reference to the checklist questions in Table-1 of Validation Protocol	Is used to summarize the responses by project participants regarding the non-conformities	Is used to summarize the responses by validation team and their conclusions

The Validation Protocol is fulfilled by the validation team in line with the descriptions above and all the CARs, CLs and FARs are listed in a transparent and clear manner.

During the validation period, a Validation Protocol which is attached in Annex 1 to this validation report was used to submit the findings to the project participants.

In line with Re Carbon Ltd. internal terminology and VCS Version 4.0, the team reports the non-conformities in the forms of Corrective Action Requests (CARs), Clarification Requests (CLs) and Forward Action Requests (FARs). When and for which type of non-conformities CARs, CLs and FARs are issued are explained below. The Validation team raises a **CAR** if one of the following occurs:

- ➢ The project participants have made mistakes that will influence the ability of the project activity to achieve real, measurable additional emission reductions
- ➢ The CDM and/or VCS requirements have not been met
- ➢ There is a risk that emission reductions cannot be monitored or calculated.

- The Validation team raises a **CL** if information is insufficient or not clear or not transparent enough to determine whether the applicable CDM and/or VCS requirements have been met.
- The Validation team raises a **FAR** during validation to highlight issues related to project implementation that require review during the first verification of the second crediting period of the project activity

According to these principles total of 04 CARs and 07 CLs were raised all of which are listed in the Validation Protocol. There hasn't been any FARs raised during the CP renewal validation.

The appointment process of the validation team takes into account the technical area(s), sectoral scope(s), and relevant host country experience required amongst team members for the accurate and thorough assessment of the project design. The relevant VCS validation and previous ITR experiences are also assessed during the selection of the team members and Independent Technical Reviewer (ITR), respectively. The validation team and ITR are assigned to this validation activity on 27/01/2022 taking all the above factors into consideration and as a result of the contract review process.

The validation team members and ITR are given in Table 2-6 below:

Table 2-6: CP renewal validation team and ITR details

Name	Role	Host Country Experience	Scope Coverage	Technical Expertise	Involvement*
Fikriye Seda Atabek	Team Leader	☒	☒	☒	A, DR, SV, R
Öykü Yakupoğlu	Trainee Validator	☒	☒	☒	A, DR, SV, R
Rohit Badaya	ITR	☒	☒	☒	ITR

* Explanations for the abbreviations used for involvement types are as follows:

A : Administrative

DR : Desk Review

SV : Online Site Visit (remote audit)

R : Reporting

ITR : Independent Technical Review

2.5.1 Forward Action Requests

The Validation team raises a FAR during the CP renewal validation to highlight issues related to project implementation that require review during the first verification of the second crediting period of the project activity as explained in the Section 2.5.

According to these principles, there hasn't been any FARs raised during the CP renewal validation.

3 VALIDATION FINDINGS

3.1.1 Project Details

KARADENİZ HES ELEKTRİK A.Ş. has installed a hydro power plant (Uzundere HEPP) in Black Sea Region of Turkey, within the province of Rize, Turkey. Uzundere HEPP involves installation and operation of a licensed project. Uzundere HEPP project entails the construction and operation of a 63.068 MWm / 62.152 MWe hydroelectric power plant. The project consists of hydro power plant with two turbines of 31.534 MWm / 31.076 MWe capacity. The project ownerships have been confirmed through the provided electricity generation licence and provisional acceptance document.

Implementation of the project consists of construction of the following main items:

• Two diversion weirs, where water from the river is diverted into conveyance pipes,

• The conveyance tunnel, which is 11.324 km,

• The powerhouse with two vertical shaft Pelton turbines, each having a capacity of 31.6 MW.

• Two generators attacted to the facility, each with a power factor of 0,9 d/d, a frequency of 50 Hz and an output of 34.5 MVA.

The net electricity production (delivered to the grid after losses and consumption in the plant) from the plant is estimated to be 156,205 MWh per annum during the second crediting period. Estimated CO2 reduction is 76,997 tonnes annually while it is calculated as 769,969 tonnes for the whole crediting period.

The project scale is "project" in line with VCS rules and requirements due to its total capacity (63.068 MWm / 62.152 MWe) and since the project is a greenfield project activity, pre-project conditions are same with the baseline scenario as discussed in the Section 3.3.4 of the validation report. The installed capacity and the expected annual generation amount of the project have been checked through the provisional acceptance document dated as 27/05/2010.

The project start date of the project activity is the initial date of selling of the produced electricity which is 27/05/2010 and the project is in compliance with the Turkish regulations. The crediting period of the project activity is renewable crediting period of 10 years 0 month which can be renewed one more time since this will be the second crediting period of the project.

The proposed project's net GHG emission reductions or removals will neither be used for compliance with an emissions trading program or to meet binding limits on GHG emissions, nor for other forms of environmental credit. The project has neither been registered, nor is seeking registration under any other GHG programs other than VCS under VER scheme.

The project has been contributing to SDG-7 through net electricity generation by renewable source, SDG-8 through employment and SDG-13 through GHG emission reduction and such actual contributions will be checked during the second crediting period.

Besides that, there hasn't been any stated commercially sensitive information by the PP at the time of CP renewal validation.

Through document review and interviews during the remote audit, Re Carbon Ltd. confirms that the description provided of the project is accurate, complete, and provides an understanding of the nature of the project.

3.2 Safeguards

3.2.1 No Net Harm

There hadn't been any observed significant environmental impact of the project activity as indicated in the registered PD and revised PD for the second crediting period and this was also confirmed through the reviewed documents in as detailed in Section 3.2.3 of the report.

3.2.2 Local Stakeholder Consultation

There hadn't been any complaint raised by the interviewed local stakeholders during the remote audit as detailed in Sections 2.3 and 2.4 of the report.

The local stakeholders as stated in the Table 2-2 above were interviewed about the following issues and there hadn't been any complaint by the interviewed local stakeholders during the remote audit:

- Noise due to the project activity
- Impact on the aquatic life where the project had been constructed
- Sufficiency of local employment (The interviewed local stakeholders were pleased about the provided local employment opportunities by the PP)
- Waste management practices implemented by PP

It was also concluded that the grievance mechanism is in place and this was also confirmed by the interviewed local stakeholders during the remote audit. The documents showing the contact details of the relevant person within PP with the signature of Çataldere and Incesu Villages Mukhtar (Village Head) and dated as 26/02/2021 were also provided to VVB.

CL-2 was raised to clarify the details on communication methods with locals. This finding had been closed as detailed in Annex-1.

3.2.3 Environmental Impact

The EIA positive decision dated as 01/04/2010 by the Provincial Directorate of Environment and Urbanization was also provided by the PP.

Life line water control is handled by the State Hydraulic Works from Regulator. The fish passage is operational within the context of the project. This is verified via photos and also interviews. Also, PP reports the minimum water amounts to State Hydraulic Works and official declaration dated 17/02/2022 is provided to VVB.

Besides that, the photos of waste storage areas and the hazardous waste records dated 2012-2017 have been provided by the PP.

Social security records have been provided for 16 employees. Besides, training records for first aid training and working with high voltage training have been provided.

CL-3 and CL-4 were raised to clarify the details on environmental impacts. These findings had been closed as detailed in Annex-1.

3.2.4 Public Comments

As confirmed during remote audit interviews, the contact information of the plant responsible exist at the Mukhtars of Çataldere and İncesu villages, the project owner and local community are always in touch. The project owner regularly checks with the Mukhtar if any complaint or a request exists. Any complaint or need from the local community could directly be received by the project owner and appropriate contributions or improvements are made to the local community. Regarding the monitoring period, signed letters by the Mukhtars of Çataldere and İncesu (dated on 26/02/2021) have been provided as declaring that the related information has been available to the villagers and they did not have any complaint.

3.3 Application of Methodology

3.3.1 Title and Reference

The project activity was earlier registered using the methodology ACM0002 version 11. The PD has been updated using the latest approved version of the methodology ACM0002 version 20. The PPs have used the most recent version of the same methodology as the original registered PD, i.e., the version that is valid at the time of submission of the revised PD for the renewal of the crediting period.

The project activity applies approved consolidated baseline and monitoring methodology "ACM0002 version 20.0: "Consolidated baseline methodology for grid-connected electricity generation from renewable sources" and the associated tools:

-
- Tool to calculate the emission factor for an electricity system, Version 07.0
- Tool to determine the remaining lifetime of equipment, Version 01
- Tool 11: Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period, Version 3.0.1

According to ACM0002 version 20.0, the latest approved tools shall be referenced in the PD like, "Tool to calculate the emission factor for an electricity system" (Version 07), "Tool to determine the remaining lifetime of equipment" (version 01), "Tool 11: Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period, Version 3.0.1" which are the latest versions of the mentioned tools valid at the starting time and the above tools are applied to the VCS-PD. Therefore, it could be concluded that the title, version and reference of the methodology including the associated tools are correct and valid.

3.3.2 Applicability

Re Carbon Ltd. has assessed the relevant information contained in the VCS-PD, remote audit and evidence obtained against the application criteria listed in the methodology.

The methodology ACM0002: Grid-connected electricity generation from renewable sources is applicable to grid-connected renewable power generation project activities that a) install a Greenfield power plant; b) involve a capacity addition to (an) existing plant(s); c) involve a retrofit of (an) existing operating plants/units; d) involve a rehabilitation of (an) existing plant(s)/unit(s); or e) involve a replacement of (an) existing plant(s)/unit(s).

The project activity installs a new power plant at a site where no renewable power plant was operated prior to the implementation of the project activity (greenfield), ACM0002: Grid connected electricity generation from renewable sources is applicable.

The applicability of this methodology is justified as below:

- The proposed project activity (Uzundere HEPP) is a greenfield, renewable (hydro power) electricity generation project, connected to the Turkish national grid,
- The project activity is the installation of hydro power plant
- The project does not involve capacity addition, a retrofit of (an) existing plant(s) or a replacement of (an) existing plant(s)
- Project activity does not involve switching from fossil fuels to renewable energy sources at the site of project activity
- The project does not involve combined heat and power generation activity
- The geographic and system boundaries for the Turkish national electricity grid can be clearly identified and information on the characteristics of the grid is available.

Additionally, the proposed project activity meets applicability criteria of TOOL07: Tool to calculate the emission factor for an electricity system; Version 7.0:

- Project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid. There is only one national grid in Türkiye.
- The emission factor for only grid power plants (off grid power plants are not taken into account) have been used by Turkish Ministry of Energy and Natural Resources to calculate emission factor.
- Tool restricts use of Tool 07 to non-annex 1 countries but that is for CDM application, this project is voluntary reduction carbon project and thus can apply the Tool 07 in Türkiye (Annex-1) country. Specially Türkiye has special circumstances and did not take any place withing compliance market.
  The calculation of the emission factor does not involve any biofuels.

Therefore, Re Carbon Ltd. confirms that the selected baseline and monitoring methodology is applicable to the project activity, and this version will be valid at the time of submitted of the project activity for CP renewal.

3.3.3 Project Boundary

As clearly defined in the revised PD, the project boundary encompasses the physical, geographical site of the renewable generation source. The hydro power plant with all installation is the project boundary and this had been validated during the remote audit.

The spatial extent of the project boundary includes the project power plant and all power plants connected physically to the electricity system that the VCS project power plant is connected to as clearly defined in the applied methodology.

As the project activity is connected to Turkish national grid, the baseline boundary is defined as the Turkish national grid. This includes the project site and all power plants connected physically to the Turkish national grid and there are no off-grid power plants in Turkish grid.

The baseline emission source is CO_2 and fossil fuels fired for electricity generation cause CO_2 emissions which is the main source. It is included to baseline calculation to find the displaced amount by the project activity. The other sources and gases are minor sources and excluded due to the simplification in line with the applied methodology.

The project activity is greenfield hydro power plant project, so there hasn't been any project emissions and sources, reservoirs and gas in line with the applied methodology.

As a result, Re Carbon Ltd. confirms that the project boundary and associated GHG sources have been clearly and correctly identified in the PD by the PP and will be valid during the second crediting period.

3.3.4 Baseline Scenario

The project activity was earlier registered using the methodology ACM0002 version 11. The PD has been updated using the latest approved version of the methodology ACM0002 version 20. All the applicability conditions of the methodology have been justified appropriately in the revised PD, version 4 and dated as 06/01/2023.

The PP has also included "Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period version 03.0.1" under the applicable tools list. The VVB has checked the application of the aforesaid tool and confirms that it has been correctly applied.

There has been no significant change in the relevant policies and circumstances, which would impact the baseline scenario since 17/03/2011 (date of initial validation) till date. The earlier registered PD takes into account all the relevant national and sectoral policies and circumstances that were applicable as on date. The discussion on the same has also been provided in the updated PD.

The project activity is supplying power to the Turkish national grid. Thus, the baseline scenario continues to remain same as earlier, as follows: "Electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the "Tool to calculate the emission factor for an electricity system".

Further, the emission factor has been updated and fixed ex-ante for the 2nd renewable crediting period. The procedures as defined in the "Tool to calculate the emission factor for an electricity system", version 07.0 have been followed. The grid emission factor (EFgrid,CM,y) in the earlier registered PD was 0.559 tCO_2/MWh whereas the grid emission factor (EFgrid,CM,y) in the updated PD is 0.4929 tCO_2/MWh with 0.25 and 0.75 weightage factor given to OM and BM respectively.

The OM is taken as 0.7258 tCO2/MWh.

The BM is taken as 0.4153 tCO2/MWh.

All other projects than wind and solar power generation project activities wOM = 0.5 and wBM = 0.5 for the first crediting period, and wOM = 0.25 and wBM = 0.75 for the second and third crediting period. Therefore wOM = 0.25 and wBM = 0.75 have been considered for the second crediting period. CM is is taken as 0.4929.

The combined margin (CM) emission factor of Turkish National Transmission System has been published by the Ministry of Energy and Natural Resources. The latest values belong to the 2019 statistics. According to this, the emission factor is taken as 0.4929.

The same has been checked from the following link and the document available: https://enerjiapi.etkb.gov.tr/Media/Dizin/ETKB/Duyurular//0c6b62ea-bf2f-4fea-b9b3-28bc6f48ddf2_Bilgi_Formu_-_Web_Sitesi.pdf

VVB confirms that the Turkish National EF has been calculated to be in line with the the “Tool to calculate the emission factor for an electricity system”, version 07.0.

No updates in policy and regulatory framework comparing with the initial validation process have been found in Turkey. Therefore, it can be concluded that the baseline scenario has not changed and continues to be the same as during the second crediting period.

3.3.5 Additionality

The project’s additionality had been checked during the initial validation of the project activity and the project was found to be as additional by the relevant VVB at that time.

It is declared in PD section 3.5 that “The project is not being implemented to comply with any law or other regulatory framework. It is a voluntary project”. Therefore, regulatory surplus has been demonstrated by PP.

Re Carbon Ltd. can also confirm that in Turkish legislation there is no law or any other regulation that mandates such power plants to be built, therefore project investment is voluntary and it can be concluded that regulatory surplus is demonstrated by PP appropriately.

3.3.6 Quantification of GHG Emission Reductions and Removals

The emission reduction calculation estimations have been included in the PD in line with the latest approved version of the methodology ACM0002 version 20.0. The baseline emissions are calculated based on the combined emission factor multiplied by the expected net electricity generation, which amounts to 76,997 ton CO_2 per annum.

Emission factor had been calculated in line with the selected methodology and the Ministry of Energy and Natural Resources document named as “Turkey’s National Electricity Network Emission Factor Factsheet, EF of hydropower plants” as 0.4929 tCO_2/MWh.

BE_y = 156,205 MWh × 0.4929 tCO_2e/MWh = 76,997 tCO_2e

There is no project emission resulting from the reservoir area of the Project Activity as the power density of the project is greater than 10W/m2. Accordingly, The Project Activity’s power density was calculated as below: =63,000,000 W/ 4,901.72m2 6 = 12,852.6 W/m2

There are no project emissions or leakage emissions associated with the hydro power project. Thus, the emission reductions correspond to the baseline emissions and the project is expected to result in an average emission reduction of 76,997 tCO_2e/year during the second crediting period.

$ER_y = BE_y – PE_y – LE_y$

$LE_y = 0$, $PE_y = 0$

$ER_y = BE_y$ =76,997 tCO_2e

CAR-1 and CAR-4 were raised regarding the emission reduction calculations during the CP renewal validation and CL-5 had been raised for demonstration of power density and they had been closed as detailed in Annex-1.

3.3.7 Methodology Deviations

No methodology deviations have been applied to the project.

3.3.8 Monitoring Plan

According to ACM0002 version 20.0, one of the parameters required to be monitored is “net electricity supplied by the proposed project to the grid in year y, $EG_{facility,y}$ which will be continuously measured and recorded at least monthly.

Net electricity generation will be based on measured value of electricity export and import and recorded via meters sealed by TEIAS (the distribution and grid company) for billing purposes therefore no new additional protocol will be needed for monitoring emission reduction. According to meter reading protocols, the internal consumption of the facility was subtracted from the gross generation. EPIAS records will be used as the main source for the quantity of net electricity delivered to the grid, and it has been cross checked with the meter reading records (OSF forms) provided to the company by TEIAS.

The site electricity technicians and plant manager will be responsible for the electricity generated, gathering all relevant data and keeping the records.

There are two electricity meters, one main meter and one back up meter. All meters are inspected and sealed by TEIAS before the commissioning of the power plant in order to be protected from interference by any of the parties and the relevant information about the electricity meters including the serial numbers have been provided by the PP. Installation of the meters and data monitoring will be carried out according to the relevant regulation by TEIAS which will record the meter readings via EPIAS system and through remote reading. The main data source will be EPIAS data and TEIAS meter records will be used for cross checking purpose.

The details about the currently available electricity meter details are as follow as in the Table 3-1 below:

Table 3-1: Electricity meter details

Model	Serial Number	Accuracy Class
EMH	Main Meter: 9674605	1s
EMH	Back-up Meter: 9674606	1s

The following meters were changed on 15.10.2020:

Model	Serial Number	Accuracy Class
ACTARIS	Main Meter: 53064430	0.5s

Model	Serial Number	Accuracy Class
ACTARIS	Back-up Meter: 53064429	0.5s

The installation and latest test date of the meters are 15.10.2020.

All data will be kept for at least two years after the crediting period for QA/QC purposes. The calibration and maintenance of the meters will be carried out in line with the “Regulation of Metering and Testing of Metering Systems”. The meters will be calibrated by TEIAS when there is an inconsistency between main and back-up meters.

Since the power density is still greater than 10W/m2, CAPPJ and APJ have not been put as monitored parameters during the second crediting period.

CAR-3 is raised regarding the monitoring during the CP renewal validation and they had been closed as detailed in Annex-1.

Besides, validation team has not identified emission sources that are not addressed by the applied methodology which are expected to contribute more than 1% of the annual emission reduction.

Therefore, Re Carbon Ltd. can confirm that the list of parameters that need to be monitored ex post for the second crediting period is complete and consistent with the relevant applied methodology which is ACM0002 version 20.0.

By document review and remote audit observations, it is also confirmed by the validation team that the monitoring plan can be properly implemented, all monitoring arrangements are feasible within the project design, and the means of implementation of the monitoring plan, including data management and quality assurance and quality control procedures, are sufficient to ensure that the emission reductions to be achieved by the project activity can be properly reported and verified.

3.4 Non-Permanence Risk Analysis

N/A (The project is not an AFOLU project).

4 VALIDATION CONCLUSION

Re Carbon Ltd. has performed the CP renewal validation of the “UZUNDERE I 63.0 MW HYDROELECTRIC POWER PLANT PROJECT, TURKEY” between 10/04/2022 and 06/01/2023. The validation was performed on the basis of VCS criteria, UNFCCC criteria for the CDM and Host Party criteria, as well as criteria given to provide for consistent project operations, monitoring and reporting.

The validation has been performed by a validation team consisting of “Fikriye Seda Atabek as team leader, Öykü Yakupoğlu as Trainee Validator and Rohit Badaya as an ITR”, and the project activity was checked against the applicable rules and regulations of CDM including CDM Validation and Verification Standard version 0.3, CDM Project Standard version 0.3, CDM Project Cycle Procedure version 0.3 and VCS Version 4.0.

Re Carbon Ltd. hereby confirms that the proposed project activity “UZUNDERE I 63.0 MW HYDROELECTRIC POWER PLANT PROJECT, TURKEY” has applied all relevant applicable guidance documents as the selected baseline and monitoring methodologies and the associated methodological tools have been applied correctly. Total emission reductions from the project are estimated to be on the average 76,997 tCO_2e per year over the selected 10 year crediting period. The emission reduction forecast has been checked and it is deemed likely that the stated amount is achieved given that the underlying assumptions do not change.

As a result, the CP renewal validation team assigned by the Re Carbon Ltd. concludes that the proposed Project Activity “UZUNDERE I 63.0 MW HYDROELECTRIC POWER PLANT PROJECT, TURKEY”, as described in the PD, version 4 and dated as 06/01/2023:

- meets all relevant Host Country criteria;
- meets all relevant requirements of the VCS project activities [including VCS Version 4.0, Article 12 of the Kyoto Protocol, the Modalities and Procedures for CDM (Marrakesh Accords) and the subsequent decisions and guidance by the COP/MOP and the CDM Executive Board];
- applies correctly the baseline and monitoring methodology ACM0002 version 20.0;
- is likely to achieve estimated emission reductions;

Therefore, Re Carbon Ltd. requests the renewal of crediting period of the project activity.

Fikriye Seda ATABEK	Rohit BADAYA	Esin TUNALI
Team Leader	ITR	Certification Manager
06/01/2023	06/01/2023	06/01/2023

ANNEX 1: VALIDATION PROTOCOL

Table 1 – VCS Project Description and CDM Validation Requirements

Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
Cover Page and General Requirements					
1. Are all items in the box at the bottom of the cover page completed using Arial 10.5 pt, black, regular (non-italic) font?	VCS Std. Version 4.0	DR	Yes, they are in Arial 10 and regular font.	OK	OK
2. Are the followings provided at the cover page in a tabular format?	VCS Std. Version 4.0	DR	Yes, they have been provided at the cover page in the tabular format.	OK	OK
2.1. Name of the project?	VCS Std. Version 4.0	DR	Yes, it has been provided at the cover page in the tabular format.	OK	OK
2.2. Version number of the VCS PD?	VCS Std. Version 4.0	DR	Yes, it has been provided at the cover page in the tabular format.	OK	OK
2.3. The date of the document issued in DD-Month-YYYY format?	VCS Std. Version 4.0	DR	Yes, it has been provided at the cover page in the tabular format.	OK	OK
2.4. Individual or entity that prepared the document?	VCS Std. Version 4.0	DR	Yes, it has been provided at the cover page in the tabular format.	OK	OK
2.5. Physical address, telephone, email, website?	VCS Std. Version 4.0	DR	Yes, it has been provided at the cover page in the tabular format.	OK	OK
3. Is this box available on the title page of the final document?	VCS Std. Version 4.0	DR	Yes, the box has been provided at the cover page.	OK	OK

*DR= Document Review, I= Interview, SV= Site Visit

Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
4. Is the PD used as a basis for validation prepared in accordance with the latest template and guidance from the VCS?	VCS Std. Version 4.0	DR	Yes, it has been prepared by using the latest template.	OK	OK
5. Are the VCS PD and other documents required under the VCS Program in English?	VCS Std. Version 4.0	DR	Yes, they have been provided in English.	OK	OK
6. Is there "Table of Contents" in the PD?	VCS Std. Version 4.0	DR	Yes, Table of Content is available in the PD.	OK	OK
7. Is the project listed on VCS project pipeline? .)	Registration and Issuance Process Procedure Version 3.8 Section 3	DR	Yes, project is listed on VCS project registry.	OK	OK
1. **PROJECT DETAILS**					
1.1. **Summary Description of the Implementation Status of the Project**					
Does section 1.1 of the VCS PD include the following?	VCS Std. Version 4.0	DR	Please see in below.	OK	OK
1.1.1.1. A summary description of the technologies/measures to be implemented by the project.	VCS Std. Version 4.0	DR	a-Please include timeline of project b-Please state turbine capacities in Mwe and MWm c-Capacities do not match with license, please explain a)	CL-1	OK
1.1.1.2. The location of the project.	VCS Std.	DR	Location provided in PD	OK	OK

*DR= Document Review, I= Interview, SV= Site Visit

Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
	Version 4.0				
1.1.1.3. An explanation of how the project is expected to generate GHG emission reductions or removals.	VCS Std. Version 4.0	DR	Yes, the explanation of how the project is expected to generate GHG emission reductions or removals has been provided.	OK	OK
1.1.1.4. A brief description of the scenario existing prior to the implementation of the project.	VCS Std. Version 4.0	DR	Yes the brief description of the scenario existing prior to the implementation of the project has been provided in Section 1.13 of the PD.	OK	OK
1.1.1.5. An estimate of annual average and total GHG emission reductions and removals.	VCS Std. Version 4.0	DR	Please use appropriate coefficients for OM and BM and revise CM and all relevant numbers in PDD	CAR-1	OK
Is the purpose of the project activity described including how it contributes to the sustainable development of the Host Party?	CDM-PDD-FORM version 11.0 EB 101 Report Annex 1 §36c	DR	Yes, the purpose of the project activity has been provided.	OK	OK
Has the VCS PD been prepared in accordance with the latest guidelines for completing the VCS PD?	VCS Std. Version 4.0	DR	Yes, the PD has been prepared by using the latest template.	OK	OK
1.2. **Sectoral Scope and Project Type**					
Is this a grouped project?	VCS Std. Version 4.0	DR	No, this isn't a grouped project.	OK	OK
Is the technical area of the project activity specified?	VCS Std. Version 4.0	DR	Yes, the sectoral scope of the project activity has been specified.	OK	OK

*DR= Document Review, I= Interview, SV= Site Visit

Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
Is the category of the project activity specified?	VCS Version 4.0	DR	Yes, the category of the project activity has been specified.	OK	OK
Is it justified how the proposed project activity conforms to the project category selected?	CDM-PDD-FORM Version 11.0 VCS Std. Version 4.0	DR	Yes, this justification has been provided in the category selected.	OK	OK
1.3. **Project Eligibility**					
Has it been described and justified how the project is eligible under the scope of the VCS Program?	VCS Version 4.0	DR	This is second CP so eligibility is not discussed once more.	OK	OK
1.4. **Project Design**					
Has it been indicated whether the project has been designed to include one of the following?	VCS Version 4.0	DR	Please see below	OK	OK
1.4.1.1. A single installation of an activity	VCS Version 4.0	DR	Yes	OK	OK
1.4.1.2. Multiple project activity instances or	VCS Version 4.0	DR	N/A	OK	OK
1.4.1.3. Grouped project.	VCS Version 4.0	DR	N/A	OK	OK
1.5. **Project Proponent**					

*DR= Document Review, I= Interview, SV= Site Visit

Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
Are the contact information for the project proponent(s) provided?	VCS Std. Version 4.0	DR	Yes	OK	OK
1.6. **Other Entities Involved in The Project**					
Are the contact information and roles/responsibilities for any other entities involved in the development of the project provided?	VCS Std. Version 4.0	DR	N/A	OK	OK
1.7. **Ownership**					
Has the evidence of project ownership been provided in accordance with the VCS Program specifications on project ownership?	VCS Std. Version 4.0	DR	stated	OK	OK
1.8. **Project Start Date**					
Is the project start date (the date on which the project began reducing or removing GHG emissions) indicated in day, month and year format?	VCS Version 4.0	DR	Yes, the project start date has been provided in the PD.	OK	OK
Is the starting date defined in accordance with the relevant guidance?	CDM-PDD-FORM Version 11.0	DR	Project start date is 27/05/2010 (The commissioning date of the project).	OK	OK
Does the VCS PD include a description on the following?	CDM-PDD-FORM Version 11.0	DR	Please see in below.	OK	OK

*DR= Document Review, I= Interview, SV= Site Visit

Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
1.8.3.1. how this starting date has been determined?	CDM-PDD-FORM Version 11.0	DR	Project start date is 27/05/2010 (The commissioning date of the project).	OK	OK
1.8.3.2. the evidence available to support this start date?	VCS Version 4.0 CDM-PDD-FORM Version 11.0	DR	Yes, the provisional acceptance Protocol has been provided.	OK	OK
1.9. **Project Crediting Period**					
Is the total crediting period including the day, month and year for the start and end dates and the total number of years indicated?	VCS Std. Version 4.0	DR	Project's second crediting period: Start date of the second crediting period: 27/05/2020 End date of the second crediting period: 26/05/2030 (both dates included). The crediting period is 10 years.	OK	OK
1.10. **Project Scale and Estimated GHG Emission Reductions or Removals**					
Is the scale of the project (project or large project) indicated?	VCS Std. Version 4.0	DR	Yes, the scale of the project has been indicated as "project"	OK	"OK
Are the estimated annual GHG emission reductions or removals during the project crediting period indicated in a tabular format?	VCS Std. Version 4.0	DR	Yes, the estimated annual amount has been indicated in the tabular format.	OK	OK

*DR= Document Review, I= Interview, SV= Site Visit

Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
1.11. **Description of the Project Activity**					
Does the proposed VCS project activity involve the alteration of an existing installation or process?	EB 101 Report Annex 2 §51	DR	N/A (This is a greenfield project.)	OK	OK
If the proposed project activity includes the alteration of an existing installation or process does the project description clearly state the differences resulting from the project activity compared to the pre-project situation?	EB 101 Report Annex 2 §51	DR	N/A (This is a greenfield project.)	OK	OK
Does the VCS PD include the following?	CDM-PDD-FORM Version 11.0	DR	Please see in below.	OK	OK
1.11.3.1. The scenario existing prior to the start of the project activity with a list of equipment and systems in operation at that time	CDM-PDD-FORM Version 11.0	DR	The scenario existing prior to the start of the project activity given in 1.13 of PD	OK	OK
1.11.3.2. The scope of activities/measures that are being implemented within the project activity, with a list of the equipment(s) and systems that will be installed and/or modified within the project activity;	CDM-PDD-FORM Version 11.0	DR	Yes, the list of equipment that will install within the project activity has been provided	OK	OK
1.11.3.3. The baseline scenario (if same as the scenario existing prior to implementation, then it should only be stated that both scenarios are the same)	CDM-PDD-FORM Version 11.0	DR	Yes, baseline scenario has been defined.	OK	OK

*DR= Document Review, I= Interview, SV= Site Visit

Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
Does the description of the scenarios include the following:	CDM-PDD-FORM Version 11.0	DR	Please see in below.	OK	OK
1.11.4.1. A list and arrangement of the main manufacturing / production technologies, systems and equipment involved	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	Yes, the list is available.	OK	OK
1.11.4.2. The emission sources and GHGs involved in the project activity, according to the methodology used	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	The emission sources and GHGs involved in the project activity, are negligible	OK	OK
1.11.4.3. The description of the functions of the equipment that is being modified and/or installed under the project activity and its relation, if any, to other manufacturing/production equipment and systems outside the project boundary	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	N/A (There hasn't been any modifications and/or installed under the project activity)	OK	OK
Is the lifetime of the project activity(s) indicated?	VCS Std. Version 4.0	DR	The economic life of an HEPP in Turkey is assumed as 50 years. However, it is recommended to renew the equipment in every 20 years.	OK	OK
Has the appropriate and clear evidence been provided regarding the expected operational lifetime of the project activity?	CDM-PDD-FORM Version 11.0	DR	Yes	OK	OK

*DR= Document Review, I= Interview, SV= Site Visit

Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
1.12. **Project Location**					
Is the location and geographical boundary of the project activity clearly identified including:	CDM-PDD-FORM Version 11.0 VCS Std. Version 4.0	DR	Please see in below.	OK	OK
1.12.1.1. Host Party(ies)?	CDM-PDD-FORM Version 11.0 VCS Std. Version 4.0	DR	Stated correctly	OK	OK
1.12.1.2. Region/State/Province etc.?	CDM-PDD-FORM Version 11.0 VCS Std. Version 4.0	DR	Stated correctly	OK	OK
1.12.1.3. City/Town/Community etc.?	CDM-PDD-FORM Version 11.0 VCS Std. Version 4.0	DR	Stated correctly	OK	OK
1.12.1.4. a set of geodetic coordinates.	VCS Std. Version 4.0	DR	Stated correctly	OK	OK
1.13. **Conditions Prior to Project Initiation**					
Are the conditions prior to project initiation provided?	VCS Std. Version 4.0	DR	Yes, the pre-project situation has been defined as baseline scenario.	OK	OK

*DR= Document Review, I= Interview, SV= Site Visit

Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
Is it demonstrated that the project has not been implemented to generate GHG emissions for the purpose of their subsequent reduction, removal or destruction?	VCS Std. Version 4.0	DR	Yes	OK	OK
1.14. **Compliance with Laws, Statutes And Other Regulatory Frameworks**					
Does the VCS PD include identification of relevant local, regional and national laws, statutes and regulatory frameworks related to the project and demonstration of compliance with them?	VCS Std. Version 4.0	DR	Yes, these have been identified.	OK	OK
1.15. **Participation Under Other GHG Programs**					
Projects Registered (or seeking registration) Under Other GHG Program(s)					
1.15.1.1. Has it been indicated whether the project has been registered, or is seeking registration under any other GHG programs in the Section 1.15.1 of the PD?	VCS Std. Version 4.0	DR	It is stated that “The project doesn’t claim Green or White certificates or equivalents that may result in double counting as a result of carbon dioxide emission reduction purposes.” A search in Gs registry, GCC and IREC reveals that project is only registered to VCS	OK	OK
1.15.1.2. If the project has been registered under any other GHG program, has the registration number and the relevant details been provided in the Section 1.15.1 of the	VCS Std. Version 4.0	DR	N/A	OK	OK

*DR= Document Review, I= Interview, SV= Site Visit

Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
PD?					
Projects Rejected by Other GHG Programs					
1.15.2.1. Has it been indicated whether the project has been rejected by any other GHG programs in the Section 1.15.2 of the PD?	VCS Std. Version 4.0	DR	N/A	OK	OK
1.15.2.2. If the project has been rejected by any other GHG programs, has the relevant information, including the reason(s) for the rejection and justification of eligibility under the VCS Program been provided in the Section 1.15.2 of the PD?	VCS Std. Version 4.0	DR	N/A	OK	OK
1.15.2.2.1. the reductions or removals generated by the project have or will not be used in the emissions trading program	VCS Std. Version 4.0	DR	N/A	OK	OK
1.15.2.2.2. for the purpose of demonstrating compliance with the binding limits that are in place in that jurisdiction or sector	VCS Std. Version 4.0	DR	N/A	OK	OK
1.16. **Other Forms of Environmental Credit**					

*DR= Document Review, I= Interview, SV= Site Visit

Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
Emissions Trading Programs and Other Binding Limits					
1.16.1.1. Has it been indicated whether the project reduces through following?	VCS Std. Version 4.0	DR	N/A	OK	OK
1.16.1.1.1. GHG emissions from activities that are included in an emissions trading program or	VCS Std. Version 4.0	DR	N/A	OK	OK
1.16.1.1.2. Any other mechanism that includes GHG allowance trading, and include details about any such programs or mechanisms	VCS Std. Version 4.0	DR	N/A	OK	OK
Other Forms of Environmental Credit					
1.16.2.1. Has it been indicated whether the project(s) created other forms of environmental credit including renewable energy certificates?	VCS Std. Version 4.0	DR	N/A	OK	OK
1.16.2.2. If the project(s) created other forms of environmental credit (for example renewable energy certificates), has the PPs provided all relevant information about the GHG-related environmental credit and the related program?	VCS Std. Version 4.0	DR	N/A	OK	OK
1.16.2.3. Have all other programs	VCS Std.	DR	N/A	OK	OK

*DR= Document Review, I= Interview, SV= Site Visit

Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
under which the project is eligible to participate (to create another form of GHG-related environmental credit) been listed?	Version 4.0				
1.17. **Additional Information Relevant to the Project**	**This section of the PD is not reviewed as the project is under validation for renewal of crediting period.**				
2. **SAFEGUARDS**					
2.1. **No Net Harm**					
Has any potential negative environmental and socio-economic impacts and the steps taken to mitigate them been summarized in the VCS PD?	VCS Std. Version 4.0	DR	No potential impacts stated in PD. Also in site visit no potential impacts determined and no complaints received from stakeholders	OK	OK
2.2. **Local Stakeholder Consultation**					
Does the process for and the outcomes from the local stakeholder consultation include the following?	VCS Std. Version 4.0	DR	Please explain the below: -procedures or methods used for engaging local stakeholders currently -mechanism for on-going communication -how comments are taken into account	CL-2	OK
2.2.1.1. The procedures or methods used for engaging local stakeholders (e.g., dates of announcements or meetings, periods during which input was	VCS Std. Version 4.0	DR	Asked above	OK	OK

*DR= Document Review, I= Interview, SV= Site Visit

Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
sought).					
2.2.1.2. The procedures or methods used for documenting the outcomes of the local stakeholder consultation	VCS Std. Version 4.0	DR	Asked above	OK	OK
2.2.1.3. The mechanism for on-going communication with local stakeholders	VCS Std. Version 4.0	DR	Asked above	OK	OK
2.2.1.4. How due account of all and any input received during the consultation has been taken.	VCS Std. Version 4.0	DR	Asked above	OK	OK
2.2.1.5. The details on any updates to the project design or justify why updates are not appropriate	VCS Std. Version 4.0	DR	Asked above	OK	OK
Is the invitation made in an open and transparent manner with use of appropriate means?	CDM-PDD-FORM Version 11.0	DR	N/A	OK	OK
Is it possible to conclude based on stakeholder meeting presentation materials that project was described in an understandable manner to the local stakeholders?	CDM-PDD-FORM Version 11.0	DR	N/A	OK	OK
Is the local stakeholder process completed before the validation?	VCS Std. Version 4.0	DR	N/A	OK	OK
2.3. **Environmental Impacts**					

*DR= Document Review, I= Interview, SV= Site Visit

Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
Have the PPs carried out an analysis of the environmental impacts of the proposed project activity including transboundary impacts?	VCS Std. Version 4.0 EB101 Report Annex 1 §92	DR	Please provide EIA not required certificate	CL-3	OK
Has a summary of the analysis of the environmental impacts of the project activity and references to all related documentation, provided under PD?	CDM-PDD-FORM version 11.0 EB101 Report Annex 1 §92	DR	No environmental impacts stated a-Please provide explanation in PD 2.3 about waste management activities on site, minimum water applications, fish passage, trainings, local employment. Please also mention the dates and details of evidences sent to DOE b- For CTP pipes please state glass fiber reinforced polyester pipes c-Sentence not clear: The only project item whose excavation activities has not started is Uzundere-I weir.	CL-4	OK
Has the PPs provided all conclusions and references to all related documentation?	EB101 Report Annex 1 §92	DR	N/A	OK	OK
Do the procedures of the Host Party require the PPs to conduct an environmental impact assessment?	EB101 Report Annex 1 §92 EB101 Report Annex 2 §126	DR	requested above	OK	OK

*DR= Document Review, I= Interview, SV= Site Visit

Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
If the procedures of the Host Party require the PPs to conduct an environmental impact assessment, has the PPs conducted an environmental impact assessment in accordance with the Host Party's procedures?	EB101 Report Annex 1 §93 EB101 Report Annex 2 §127 §129	DR	requested above	OK	OK
2.4. **Public Comments**					
Has it been demonstrated in the VCS PD how due account of all and any comments received during the public comment period had been taken?	VCS Std. Version 4.0	DR	a) Yes	OK	OK
Have the details on any updates to the project design or demonstrate the insignificance or irrelevance of comments been included in the VCS PD?	VCS Std. Version 4.0		Yes	OK	OK
3. **APPLICATION of METHODOLOGY**					
3.1. **Title and Reference of Methodology**					
Does the project apply the correct and valid version of the approved methodology and referred tools at the time of submission for crediting	VCS Std. Version 4.0	DR	The approved methodology with the correct and valid version has been included in the PD.	OK	OK

*DR= Document Review, I= Interview, SV= Site Visit

Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
period renewal?					
Does the VCS PD indicate the title and version of the methodology and the related tool(s) correctly?	VCS Std. Version 4.0	DR	The approved tools with the correct and valid version has been included in the PD.	OK	OK
3.2. **Applicability of Methodology**					
Is the choice of the methodology justified by showing that the proposed project activity meets all the applicability conditions of the methodology?	VCS Std. Version 4.0 CDM-PDD-FORM version 11.0 EB 101 Report Annex 1 §54 EB 101 Report Annex 2 §67	DR	Yes, applicability is discussed correctly.	OK	OK
Does the project activity meet each of the applicability conditions of the tools or other methodology components referred to in the applied methodology?	VCS Std. Version 4.0 EB 101 Report Annex 2 §67	DR	Yes	OK	OK
ACM 0002					

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
Is the type of proposed project activity defined? • ○ ○ ○ ○ ○ • •	ACM 0002 Version 20.0	Dr	Project is hydropower plant project.	OK	OK
If the proposed project activity is a hydro power plant project, does one of the following conditions conform to the proposed project activity?	ACM 0002 Version 20.0	DR		OK	OK
3.2.4.1. Is the proposed project activity implemented in an existing single or multiple reservoirs, with no change in the volume of any of the reservoirs?	ACM 0002 Version 20.0	DR	N/A	OK	OK
3.2.4.2. Is the project activity implemented in an existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density calculated using equation (3), is greater than 4 W/m2?	ACM 0002 Version 20.0	DR	N/A	OK	OK

*DR= Document Review, I= Interview, SV= Site Visit

Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
3.2.4.3. Is the project activity results in new single or multiple reservoirs and the power density calculated using equation (3), is greater than 4 W/m2?	ACM 0002 Version 20.0	DR	Please calculate and show power density in excel and PDD	CL-5	OK
If the project activity is an integrated hydro power project, has the PPs demonstrated that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project?	ACM 0002 Version 20.0	DR	N/A	OK	OK
If the project activity is an integrated hydro power project, has the PPs provided an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs?	ACM 0002 Version 20.0	DR	N/A	OK	OK
If the project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs calculated using equation (3) is lower than or equal to 4 W/m2, do all the following conditions conform the project activity?	ACM 0002 Version 20.0	DR	N/A	OK	OK

*DR= Document Review, I= Interview, SV= Site Visit

Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
3.2.7.1. The power density calculated using the total installed capacity of the integrated project, as per equation (4), is greater than 4 W/m2;	ACM 0002 Version 20.0	DR	N/A	OK	OK
3.2.7.2. Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity;	ACM 0002 Version 20.0	DR	N/A	OK	OK
3.2.7.3. Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m2 shall be:	ACM 0002 Version 20.0	DR	N/A	OK	OK
3.2.7.3.1. Lower than or equal to 15 MW; and	ACM 0002 Version 20.0	DR	N/A	OK	OK
3.2.7.3.2. Less than 10 per cent of the total installed capacity of integrated hydro power project.	ACM 0002 Version 20.0	DR	N/A	OK	OK
ACM 0001					
Does the project activity include one of the following conditions?	ACM 0001 Version 19.0	DR	N/A	OK	OK
3.2.8.1. Install a new landfill gas (LFG) capture system in an existing or new (Greenfield) SWDS where no LFG capture system was or would have	ACM 0001 Version 19.0	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
been installed prior to the implementation of the project activity; or					
3.2.8.2. Make an investment into an existing LFG capture system to increase the recovery rate or change the use of the captured LFG, provided that:	ACM 0001 Version 19.0	DR	N/A	OK	OK
3.2.8.3. The captured LFG was vented or flared and not used prior to the implementation of the project activity; and	ACM 0001 Version 19.0	DR	N/A	OK	OK
3.2.8.4. In the case of an existing active LFG capture system for which the amount of LFG cannot be collected separately from the project system after the implementation of the project activity and its efficiency is not impacted on by the project system: historical data on the amount of LFG capture and flared is available;	ACM 0001 Version 19.0	DR	N/A	OK	OK
3.2.8.5. Flare the LFG and/or use the captured LFG in any (combination) of the following ways:	ACM 0001 Version 19.0	DR	N/A	OK	OK
3.2.8.6. Generating electricity;	ACM 0001 Version 19.0	DR	N/A	OK	OK
3.2.8.7. Generating heat in a boiler, air heater or kiln (brick	ACM 0001	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
firing only) or glass melting furnace; and/or	Version 19.0				
3.2.8.8. Supplying the LFG to consumers through a natural gas distribution network;	ACM 0001 Version 19.0	DR	N/A	OK	OK
3.2.8.9. Supplying compressed/liquefied LFG to consumers using trucks;	ACM 0001 Version 19.0	DR	N/A	OK	OK
3.2.8.10. Supplying the LFG to consumers through a dedicated pipeline;	ACM 0001 Version 19.0	DR	N/A	OK	OK
3.2.8.11. Do not reduce the amount of organic waste that would be recycled in the absence of the project activity. • • • •	ACM 0001 Version 19.0	DR	N/A	OK	OK
AM0058					
Is this a project activity that introduce a primary district heating system to supply heat to residential and commercial consumers?	AM0058 Version 5.0	DR	N/A	OK	OK
If this is a project activity that introduce a primary district heating system to	AM0058	DR	N/A	OK	OK

*DR= Document Review, I= Interview, SV= Site Visit

Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
supply heat to residential and commercial consumers, does the heat come from one of the following?	Version 5.0				
3.2.10.1. An existing grid connected thermal power plant with no steam extraction for heating purposes, other than that required for the operation of the power plant auxiliary systems, prior to the project activity;	AM0058 Version 5.0	DR	N/A	OK	OK
3.2.10.2. A new centralised heat only boiler(s); or	AM0058 Version 5.0	DR	N/A	OK	OK
3.2.10.3. A combination of both 3.2.10.1 and 3.2.10.2	AM0058 Version 5.0	DR	N/A	OK	OK
Does the project activity include any of below components?	AM0058 Version 5.0	DR	N/A	OK	OK
3.2.11.1. Heat supplied to the district heating system is predominantly used for heating and/or hot tap water supply for residential and/or commercial users. At the most 20 per cent of the heat may be supplied to other users, such as for industrial production processes;	AM0058 Version 5.0	DR	N/A	OK	OK
3.2.11.2. For project activities in which a co-generation plant supplies heat to the district heating system:	AM0058 Version 5.0	DR	N/A	OK	OK

*DR= Document Review, I= Interview, SV= Site Visit

Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
3.2.11.3. The power plant is fossil fuel fired;	AM0058 Version 5.0	DR	N/A	OK	OK
3.2.11.4. Only one type of fuel is used by the project's co-generation plant (a maximum of 1 per cent of auxiliary fuel may be used for start-up.). The same type of fossil fuel is fired in the power plant in the baseline and project scenarios;	AM0058 Version 5.0	DR	N/A	OK	OK
3.2.11.5. The project activity does not lead to an increase in the technical lifetime of the power plant and does not result in any major integrated production changes at the power plant, other than the modifications required for heat extraction for the district heating.	AM0058 Version 5.0	DR	N/A	OK	OK
3.2.11.6. Emission reductions resulting from heat supply to new residential areas, in cases where more than 50% of the annual heat production originates from heat-only boilers and less than 50% of heat comes from the power plant within the primary district heating system;	AM0058 Version 5.0	DR	N/A	OK	OK
3.2.11.7. Emission reductions resulting from a decrease in heat losses due to the water losses or from demand-side measures (e.g. insulation of	AM0058 Version 5.0	DR	N/A	OK	OK

*DR= Document Review, I= Interview, SV= Site Visit

Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
buildings, use of thermostatic valves, behavioural changes due to billing practices).					
AMS-I.D.					
Does the proposed project activity comprises renewable energy units such as photovoltaic, hydro, tidal/wave, wind, geothermal and renewable biomass, supplying one of the following? • • • • • • •	AMS I.D. Version 18.0 §2 §4 §7	DR	N/A	OK	OK
3.2.12.1. Electricity to a national or a regional grid, or	AMS I.D. Version 18.0 §2 §4 §7	DR	N/A	OK	OK
3.2.12.2. Electricity to an identified consumer facility via national/regional grid through a contractual arrangement such as wheeling?	AMS I.D. Version 18.0 §2 §4 §7	DR	N/A	OK	OK
Does the new unit (proposed project activity) have both renewable and	AMS I.D. Version	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
non-renewable components?	18.0 §6 §11				
Does the new unit co-fires fossil fuel?	AMS I.D. Version 18.0 §6 §7	DR	N/A	OK	OK
Does the proposed project activity involve the addition of renewable energy generation units at an existing renewable power generation facility?	AMS I.D. Version 18.0 §8	DR	N/A	OK	OK
Is the project activity a retrofit, rehabilitation or a replacement?	AMS I.D. Version 18.0 §9	DR	N/A	OK	OK
If the proposed project activity is a hydro power plant project, does one of the following conditions conform to the proposed project activity?	AMS I.D. Version 18.0 §5	DR	N/A	OK	OK
Is the proposed project activity implemented in an existing reservoir, with no change in the volume of reservoir?	AMS I.D. Version 18.0 §5	DR	N/A	OK	OK
Is the project activity implemented in	AMS I.D.	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per the definitions given in the project emissions section, is greater than 4 W/m^2?	Version 18.0 §5				
Is the project activity results in new reservoirs and the power density of the power plant, as per the definitions given in the project emissions section, is greater than 4 W/m2?	AMS I.D. Version 18.0 §5	DR	N/A	OK	OK
AMS I.C.					
Does the proposed project activity comprise renewable energy technologies that supply users (residential, industrial or commercial facilities) with thermal energy that displaces fossil fuel use?	AMS I.C. Version 21 §2	DR	N/A	OK	OK
Is the proposed project activity a biomass based co-generation or trigeneration system?	AMS I.C. Version 21 §3 §22-a §22-b	DR	N/A	OK	OK
If the proposed project activity is a biomass based co-generation system, does the proposed project activity produce heat and power in separate element processes?	AMS I.C. Version 21	DR	N/A	OK	OK

*DR= Document Review, I= Interview, SV= Site Visit

Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
	§22-a				
If the proposed project activity is a biomass based trigeneration system, does the proposed project activity produce electrical energy and thermal energy in separate element processes?	AMS I.C. Version 21 §22-b §22-d	DR	N/A	OK	OK
If the proposed project activity is a biomass based co-generation system, does the emission reductions accrue from one of the following activities?	AMS I.C. Version 21 §4	DR	N/A	OK	OK
3.2.25.1. Electricity supply to a grid	AMS I.C. Version 21 §4-a	DR	N/A	OK	OK
3.2.25.2. Electricity and/or thermal energy (steam or heat) production for on-site consumption or for consumption by other facilities,	AMS I.C. Version 21 §4-b	DR	N/A	OK	OK
3.2.25.3. Combination of both	AMS I.C. Version 21 §4-c	DR	N/A	OK	OK
If the proposed project activity is a biomass based co-generation or trigeneration system, is the total capacity of the project activity remain	AMS I.C. Version 21 §9	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
under the below stated limits?					
3.2.26.1. 45 MW thermal if the project activity includes emission reductions from both the thermal and electrical energy components	AMS I.C. Version 21 §9-a	DR	N/A	OK	OK
3.2.26.2. 45 MW thermal if the emission reductions are solely on account of thermal energy production (i.e. no emission reductions accrue from the electricity component)	AMS I.C. Version 21 §9-b	DR	N/A	OK	OK
3.2.26.3. 15 MW if the emission reductions are solely on account of electrical energy production (i.e. no emission reductions accrue from the thermal energy component),	AMS I.C. Version 21 §9-c	DR	N/A	OK	OK
Is the total installed/rated thermal energy generation capacity of the project equipment equal to or less than 45 MW thermal?	AMS I.C. Version 21 §7	DR	N/A	OK	OK
Is the project activity a co-fired system?	AMS I.C. Version 21 §8 §22-e	DR	N/A	OK	OK
If the proposed project activity is a co-fired system, does the total installed thermal energy generation capacity of the project equipment,	AMS I.C. Version 21	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
when using both fossil and renewable fuel, exceed 45 MW thermal?	§8				
Does the project activity involve the addition of renewable energy units at an existing renewable energy facility?	AMS I.C. Version 21 §10	DR	N/A	OK	OK
If the project activity involve the addition of renewable energy units at an existing renewable energy facility, does the total capacity of the units added remain under the limits (45 MW thermal or 15 MWelecrical) and are the units added physically distinct from the existing units?	AMS I.C. Version 21 §10	DR	N/A	OK	OK
If the project activity uses solid biomass fuel, (e.g. briquette), is it demonstrated that it has been produced using solely renewable biomass?	AMS I.C. Version 21 §11 §17	DR	N/A	OK	OK
If the producer of the solid biomass fuel is not a project participant, are the project participant and the producer bound by a contract that shall enable the project participant to monitor the source of the renewable	AMS I.C. Version 21 §12	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
biomass to account for any emissions associated with solid biomass fuel production?					
If the electricity and/or steam/heat produced by the project activity is delivered to a third party, is there a contract between the supplier and consumer(s) of the energy will have to that ensures there is no double-counting of emission reductions?	AMS I.C. Version 21 §13	DR	N/A	OK	OK
If the project activity recovers and utilizes biogas for power/heat production, does it apply this methodology on a stand alone basis?	AMS I.C. Version 21 §14	DR	N/A	OK	OK
If the project activity recovers and utilizes biogas for power/heat production and applies this methodology on a stand alone basis, has the PPs taken into account any incremental emissions occurring due to the implementation of the project activity, as project or leakage emissions?	AMS I.C. Version 21 §14	DR	N/A	OK	OK
If project equipment contains refrigerants, is the refrigerant used in the project case any ozone depleting potential?	AMS I.C. Version 21	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
	§15				
If the proposed project activity is a charcoal based biomass energy generation, is the charchoal produced from renewable biomass sources?	AMS I.C. Version 21 §16	DR	N/A	OK	OK
If the proposed project activity is a charcoal based biomass energy generation, is the charcoal produced in kilns equipped with methane recovery and destruction facility?	AMS I.C. Version 21 §16-a	DR	N/A	OK	OK
If charcoal is produced in kilns not equipped with a methane recovery and destruction facility, has the PPs considered the methane emissions from the production of charcoal?	AMS I.C. Version 21 §16-b	DR	N/A	OK	OK
3.3. **Project Boundary**					
Is the diagram or map of the project boundary, showing clearly the physical locations of the various installations or management activities taking place as part of the project activity given?	VCS Std. Version 4.0	DR	Yes, the simple diagram showing clearly the physical locations of the various installations has been provided in the PD. Please revise Erdemli transformer	CAR-2	OK
Has the PP described the emission sources and GHGs included in the project boundary for the purpose of calculating project emissions and	VCS Std. Version 4.0 CDM-PDD-	DR	tabular format provided	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
baseline emissions, in the tabular format?	FORM Version 11.0				
Has the PP presented a flow diagram of the project boundary, physically delineating the project activity, based on the description provided in section 1.8 of the PD?	CDM-PDD-FORM Version 11.0	DR	Yes, the simple diagram showing clearly the physical locations of the various installations has been provided in the PD.	OK	OK
Has the PP included in the flow diagram the equipment, systems and flows of mass and energy described in PD	CDM-PDD-FORM Version 11.0 VCS Std. Version 4.0	DR	Yes, the simple diagram showing clearly the physical locations of the various installations has been provided in the PD.	OK	OK
Has it been indicated in the diagram the emission sources and GHGs included in the project boundary?	CDM-PDD-FORM Version 11.0 VCS Std. Version 4.0	DR	N/A	OK	OK
Does the selected methodology allow the PPs to choose whether a source or gas is to be included in the project boundary?	EB 101 Report Annex 1 §58	DR	Yes and gases are correctly selected	OK	OK
If the selected methodology allows the project participants to choose whether a source or gas is to be included in the project boundary, do the project participants explain and justify their choices?	EB 101 Report Annex 1 §58	DR	Yes and gases are correctly selected	OK	OK
Have all sources and GHGs necessary for the calculation of emissions been	EB 101	DR	Yes and gases are correctly selected	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
included within the project boundary?	Report Annex 2 §69				
Does the PD correctly describe the project boundary and the physical delineation of the proposed project activity?	EB 101 Report Annex 1 §57	DR	Yes	OK	OK
Has the selected methodology been correctly applied with respect to project boundary?	EB 101 Report Annex 2 §63a	DR	Yes	OK	OK
ACM 0002					
Is the spatial extent of the project boundary identified correctly?	ACM 0002 Version 20.0	DR	Yes	OK	OK
Are the greenhouse gases and emission sources included in or excluded from the project boundary given in the tabular form as per the guidance given in Table-2 of ACM 0002?	ACM 0002 Version 20.0	DR	Yes	OK	OK
ACM 0001					
Does the project boundary include the following as applicable?	ACM 0001 Version 19.0	DR	N/A	OK	OK
3.3.13.1. Sites where the LFG is flared or used (e.g. flare, power	ACM 0001 Version	DR	N/A	OK	OK

*DR= Document Review, I= Interview, SV= Site Visit

Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
plant, boiler, air heater, glass melting furnace, kiln, natural gas distribution network, dedicated pipeline or biogas processing facility);	19.0				
3.3.13.2. Captive power plant(s) (including emergency diesel generators) or power generation sources connected to the grid, which are supplying electricity to the project activity;	ACM 0001 Version 19.0	DR	N/A	OK	OK
3.3.13.3. Captive power plant(s) (including emergency diesel generators) or power generation sources connected to the grid, which are supplying electricity in the baseline that is displaced by electricity generated by captured LFG in the project activity;	ACM 0001 Version 19.0	DR	N/A	OK	OK
3.3.13.4. Heat generation equipment or sources which are supplying heat in the baseline that is displaced by heat generated by captured LFG in the project activity; and	ACM 0001 Version 19.0	DR	N/A	OK	OK
3.3.13.5. The transportation of the compressed/liquefied LFG from the biogas processing facility to consumers.	ACM 0001 Version 19.0	DR	N/A	OK	OK
AM0058					
Does the physical delineation of the project boundary include the	AM0058	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
following?	Version 5.0				
3.3.14.1. For project activities in which a power plant supplies heat to the district heating network, the site of the power plant, including the heat extraction unit(s) and all interrelated production units to account for emissions resulting from changes in power generation and consumption due to the project activity;	AM0058 Version 5.0	DR	N/A	OK	OK
3.3.14.2. The heat-only boilers that supply heat to the district heating system;	AM0058 Version 5.0	DR	N/A	OK	OK
3.3.14.3. The district heating system, including pipes, sub-stations and buildings that are or will be connected to the district heating system.	AM0058 Version 5.0	DR	N/A	OK	OK
Has it been illustrated by PP how the project boundary is defined and where the points to measure heat supplied to buildings (Qe and Qn) should be located in line with the Figure-1 in AM0058?	AM0058 Version 5.0	DR	N/A	OK	OK
Are the emissions sources included in or excluded from the project boundary indicated in the PD in line with the Table-2 of the methodology?	AM0058 Version 5.0	DR	N/A	OK	OK
AMS-I.D.					

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
Is the spatial extent of the project boundary identified correctly?	AMS I.D. Version 18.0 §18	DR	N/A	OK	OK
AMS I.C.					
Does the spatial extent of the project boundary encompass the following?	AMS I.C. Version 21 §24	DR	N/A	OK	OK
3.3.18.1. All plants generating electricity and/or thermal energy located at the project site, whether fired with biomass, fossil fuels or a combination of both	AMS I.C. Version 21 §24-a	DR	N/A	OK	OK
3.3.18.2. All power plants connected physically to the electricity system (grid) that the project plant is connected to	AMS I.C. Version 21 §24-b	DR	N/A	OK	OK
3.3.18.3. Industrial, commercial or residential facility, or facilities, consuming energy generated by the system and the processes or equipment affected by the project activity	AMS I.C. Version 21 §24-c	DR	N/A	OK	OK
3.3.18.4. The processing plant of biomass residues, for project activities using solid biomass fuel (e.g. briquette), unless all associated emissions are accounted for as leakage	AMS I.C. Version 21 §24-d	DR	N/A	OK	OK

*DR= Document Review, I= Interview, SV= Site Visit

Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
emissions or are part of an independently registered CDM project;					
3.3.18.5. The geographic boundaries of the dedicated plantations if the feedstock is biomass produced in dedicated plantations;	AMS I.C. Version 21 §24-e	DR	N/A	OK	OK
3.3.18.6. The transportation itineraries, if the biomass is transported over distances greater than 200 kilometres, unless all associated emissions are accounted for as leakage emissions;	AMS I.C. Version 21 §24-f	DR	N/A	OK	OK
3.3.18.7. The site of the anaerobic digester in the case of project activity that recovers and utilizes biogas for producing electricity and/or thermal energy and applies this methodology on a stand-alone basis, i.e. without using a Type III component of a SSC methodology.	AMS I.C. Version 21 §24-g	DR	N/A	OK	OK
3.4. **Baseline Scenario**					
Does the approved methodology that is selected by the proposed project activity prescribe the baseline scenario and hence no further analysis is required?	EB 101 Report Annex 2 §94 EB 101 Report	DR	Methodology and PD states the baseline scenario as “the electricity delivered to the grid by the project activity that otherwise would have been generated by the operation of grid-connected power plants and by the addition of new generation sources”.	OK	OK

*DR= Document Review, I= Interview, SV= Site Visit

Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
• •	Annex 1 §59				
Does the PD identify the baseline for the proposed project activity, defined as the scenario that reasonably represents the anthropogenic emissions by sources of GHGs that would occur in the absence of the proposed project activity?	EB 101 Report Annex 2 §75 EB 101 Report Annex 1 §61	DR	Yes, the baseline scenario has been identified.	OK	OK
If the methodology requires use of the tools to identify the baseline scenario, have all those been applied?	EB 101 Report Annex 2 §77	DR	N/A	OK	OK
Are there relevant national and/or sectoral policies to identify the baseline scenario?	EB 101 Report Annex 2 §81 EB 101 Report Annex 1 §64	DR	N/A	OK	OK
If there are relevant national and/or sectoral policies to identify the baseline scenario, have those been considered correctly in the PDD?	EB 101 Report Annex 2 §83d	DR	N/A	OK	OK
Are there relevant circumstances to identify the baseline scenario?	EB 101 Report Annex 2	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
	§81				
Does the methodology require several alternative scenarios to be considered in the identification of the most reasonable baseline scenario?	EB 101 Report Annex 2 §78	DR	N/A	OK	OK
If the methodology requires several alternative scenarios to be considered in the identification of the most reasonable baseline scenario, are all credible scenarios that are in the PD and are supplementary to those required by the methodology reasonable in the context of the proposed project activity?	EB 101 Report Annex 2 §78	DR	N/A	OK	OK
If the proposed project activity includes several different facilities, technologies, outputs or services, do the alternative scenarios for each of them be identified separately?	EB70 Report Annex 8	DR	N/A	OK	OK
If the alternative scenarios for each of them be identified separately, are the realistic combinations of these be considered as possible alternative scenarios to the proposed project activity?	EB70 Report Annex 8	DR	N/A	OK	OK
Does the list of alternative scenarios given in the PD include the following?	EB 101 Report Annex 2 §93	DR	N/A	OK	OK
3.4.11.1. The project activity is	EB 101	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
undertaken without being registered as a CDM project activity	Report Annex 2 §93a				
3.4.11.2. All plausible alternatives	EB 101 Report Annex 2 §93b	DR	N/A	OK	OK
3.4.11.3. Comply with all applicable and enforced legislation	EB 101 Report Annex 2 §93c	DR	N/A	OK	OK
Has the PP explained how the baseline scenario is established in accordance with the selected methodology(ies)?	CDM-PDD-FORM Version 11.0 EB93 Report Annex 4 §62	DR	Yes, this has been explained.	OK	OK
Where the procedure in the selected methodology(ies) involves several steps, has the PPs described how each step is applied and transparently documented the outcome of each step?	VCS Std. Version 4.0 CDM-PDD-FORM version 11.0 EB 101 Report Annex 1	DR	Yes	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
	§59				
Has the PP provided and explained all data used to establish the baseline scenario (variables, parameters, data sources, etc.)?	CDM-PDD-FORM version 11.0	DR	N/A	OK	OK
Is the identified baseline scenario reasonably supported by correct and verifiable references, assumptions, calculations and ratinonales?	VCS Std. Version 4.0 CDM-PDD-FORM version 11.0	DR	Yes	OK	OK
Has the selected methodology been correctly applied with respect to baseline identification?	CDM-PDD-FORM version 11.0 EB 101 Report Annex 2 §80	DR	Yes, the baseline identification has been correctly applied with the selected methodology.	OK	OK
ACM 0002					
If the project activity involves the installation of a greenfield power plant, is the baseline scenario identified appropriately in accordance with the ACM 0002?	ACM 0002 Version 20.0	DR	Methodology and PD states the baseline scenario as “the electricity delivered to the grid by the project activity that otherwise would have been generated by the operation of grid-connected power plants and by the addition of new generation sources”.	OK	OK
If the project activity involves capacity addition to existing grid-connected	ACM 0002 Version	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
renewable power plant/unit, is the baseline scenario identified appropriately in accordance with the ACM0002?	20.0				
If the proposed project activity is a capacity addtion, retrofit, rehabilitation or replacement, have the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit or rehabilitation of the plant has been undertaken between the start of this minimum historical reference period and the implementation of the project activity?	ACM 0002 Version 20.0	DR	N/A	OK	OK
If the project activity is the retrofit or replacement of existing grid-connected renewable power plant/unit, is the point of time at which the generation facility would likely be replaced or retrofitted ($DATE_{Baseline\ Retrofit}$) defined?	ACM 0002 Version 20.0	DR	N/A	OK	OK
If the project activity is the retrofit or replacement of existing grid-connected renewable power plant/unit, is the baseline scenario identified following the step-wise	ACM 0002 Version 20.0	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
procedure in accordance with the ACM0002?					
Are the realistic and credible alternative baseline scenarios for power generation appropriately identified following the Step 1 of the "Combined tool to identify the baseline scenario and demonstrate additionality"?	ACM 0002 Version 20.0	DR	N/A	OK	OK
Is "the proposed project activity undertaken without being registered as a CDM project activity" listed as one of the alternatives?	EB70 Report Annex 8 EB 101 Report Annex 2 §93a ACM 0002 Version 20.0	DR	N/A	OK	OK
Has "other realistic and credible alternative scenario(s) to the proposed CDM project activity scenario that deliver outputs services or services with comparable quality, properties and application areas" been listed as an alternative?	EB70 Report Annex 8 EB 101 Report Annex 2 §93b ACM 0002 Version 20.0	DR	N/A	OK	OK
Has "continuation of the current situation (no project activity or other	EB70	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
alternatives undertaken" been listed as an alternative?	Report Annex 8 ACM 0002 Version 20.0				
If the barrier analysis is used, is the Step 2 of the latest applicable version of "Combined tool to identify the baseline scenario and demonstrate additionality" applied appropriately?	ACM 0002 Version 20.0	DR	N/A	OK	OK
If more than one alternative is remaining after Step 2 and if the remaining alternatives include scenarios P1 and P3, is the Investment Comparison as per step 3 of the "Combined tool to identify the baseline scenario and demonstrate additionality" applied appropriately?	ACM 0002 Version 20.0	DR	N/A	OK	OK
If more than one alternative is remaining after Step 2 and if the remaining alternatives include scenarios P1 and P2, is the Benchmark Analysis as per step 2b of the "Tool for the demonstration and assessment of additionality" applied appropriately?	ACM 0002 Version 20.0	DR	N/A	OK	OK
ACM 0001					
Has the the most plausible baseline scenario been determined according to the simplified procedures or the procedures according to the latest	ACM 0001 Version 19.0	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
applicable version of the "Combined tool to identify the baseline scenario and demonstrate additionality"? a) b) c)					
AM0058					
Is the most plausible baseline scenario "no implementation of primary district heating system (continuation of current practice)"?	AM0058 Version 5.0	DR	N/A	OK	OK
AMS I.D.					
If the project activity is greenfield power plant, is the baseline scenario identified as "the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid?	AMS I.D. Version 18.0 §19	DR	N/A	OK	OK
If the project activity involves retrofits, rehabilitations or replacements of an existing facility, is baseline scenario identified appropriately in accordance with AMS I.D.?	AMS I.D. Version 18.0 §20	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
Have the PPs demonstrated the remaining lifetime of the equipment replaced according to the requirements described in the general guidelines to SSC CDM methodologies?	AMS I.D. Version 18.0 §21	DR	N/A	OK	OK
If the project activity involves capacity addition to existing grid-connected renewable energy power plant/unit,, is baseline scenario identified appropriately in accordance with AMS I.D.?	AMS I.D. Version 18.0 §21	DR	N/A	OK	OK
Have the PPs explained and documented the quantities and types of biomass and the biomass to fossil fuel ratio (in case of co-fired system) to be used during the crediting period in the PDD?	AMS I.D. Version 18.0 §44	DR	N/A	OK	OK
AMS I.C.					
If the project activity is a renewable energy technology that displaces technologies using fossil fuels, is the baseline scenario identified appropriately in accordance with AMS I.C.?	AMS I.C. Version 21 §24	DR	N/A	OK	OK
If the project activity is implemented on existing facilities, are the baseline calculations based on operational data on energy use (e.g. electricity,	AMS I.C. Version 21 §25 §26	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
fossil fuel) and plant output (e.g. steam/chilling/electricity thermal and/or electrical energy) using one of the following?					
3.4.37.1. Most recent three years operational data immediately prior to the start date of the project activity in case of existing facilities which has more than three years of operation history;	AMS I.C. Version 21 §26-a	DR	N/A	OK	OK
3.4.37.2. A minimum of most recent one year data in case of existing facilities which has more than three years of operation history but do not have three years operational data; or	AMS I.C. Version 21 §26-b	DR	N/A	OK	OK
3.4.37.3. A performance test/measurement campaign carried out prior to the implementation of the project activity in case of existing facilities which has more than three years of operation history but do not have operational data/information such as efficiency, energy consumption and output or the available data is not reliable due to various factors such as the use of imprecise or non-calibrated	AMS I.C. Version 21 §26-c	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
measuring equipment.					
If the project activity is implemented in existing faclities where the additionality is demonstrated based on a baseline scenario that is not the continuation of the current practice, has the PPs chosen the baseline emission factor as the lower of the following two choices?	AMS I.C. Version 21 §28	DR	N/A	OK	OK
3.4.38.1. The emission factor of the fossil fuel that would have been used in the absence of the project activity;	AMS I.C. Version 21 §28-a	DR	N/A	OK	OK
3.4.38.2. The emission factor of the fossil fuel that was used prior to the implementation of the project activity.	AMS I.C. Version 21 §28-b	DR	N/A	OK	OK
If the proposed project activity is producing both heat and thermal energy, is the baseline scenario determined as one of the following?	AMS I.C. Version 21 §29	DR	N/A	OK	OK
3.4.39.1. Electricity is imported from a grid and thermal energy (steam/heat) is produced using fossil fuel	AMS I.C. Version 21 §29-a	DR	N/A	OK	OK
3.4.39.2. Electricity is produced in an on-site captive power plant using fossil fuel (with a possibility of export to the grid) and thermal energy is produced using fossil fuel	AMS I.C. Version 21 §29-b	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
3.4.39.3. A combination of 3.4.39.1 and 3.4.39.2	AMS I.C. Version 21 §29-c	DR	N/A	OK	OK
3.4.39.4. Electricity and thermal energy) are produced in a cogeneration or trigeneration unit using fossil fuel (with a possibility of export of electricity to a grid/other facilities and/or thermal energy to other facilities);	AMS I.C. Version 21 §29-d	DR	N/A	OK	OK
3.4.39.5. Electricity is imported from a grid and/or produced in an on-site captive power plant using fossil fuels (with a possibility of export to the grid); steam/heat is produced using biomass	AMS I.C. Version 21 §29-e	DR	N/A	OK	OK
3.4.39.6. Electricity is produced in an on-site captive power plant using biomass (with a possibility of export to a grid) and/or imported from a grid; thermal energy is produced using fossil fuel	AMS I.C. Version 21 §29-f	DR	N/A	OK	OK
3.4.39.7. Electricity and thermal energy are produced in a biomass fired cogeneration or trigeneration unit (without a possibility of export of electricity either to a grid or to other facilities and without a possibility of export of thermal	AMS I.C. Version 21 §29-g §30	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
energy to other facilities)					
3.4.39.8. Electricity and/or thermal energy produced in a co-fired system	AMS I.C. Version 21 §29-h	DR	N/A	OK	OK
3.4.39.9. Electricity is imported from a grid and/or produced in a biomass fired cogeneration or trigeneration unit (without a possibility of export of electricity either to the grid or to other facilities); thermal energy is produced in a biomass fired cogeneration or trigeneration unit and/or a biomass fired boiler (without a possibility of export of thermal energy to other facilities)	AMS I.C. Version 21 §29-i §31	DR	N/A	OK	OK
3.4.39.10. Electricity is imported from a grid and/or produced in an on-site captive power plant using fossil fuel and thermal energy is produced using electricity.	AMS I.C. Version 21 §29-j	DR	N/A	OK	OK
If the project activity seeks to retrofit or modify an existing facility for renewable energy generation, is the baseline scenario determined	AMS I.C. Version 21 §58 §59	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
according to the methodology?					
3.5. **Additionality**	**This section of the PD is not reviewed as the project is under validation for renewal of crediting period but please check the regulatory surplus as in below.**				
Has the regulatory surplus been demonstrated in accordance with the latest applicable requirements set out in the VCS Program available at the time of crediing period renewal process?	VCS Std. Version 4.0 Validation and Verification Manual Version 3.2	DR	Please confirm in the Section 3.5 of the PD that project is not being implemented to comply with any law or other regulatory framework.	CL-6	OK
3.6. **Methodology Deviations**					
If there are deviations from the methodology, has the PPs provided and justified the methodology deviations?	VCS Std. Version 4.0	DR	The most current version of ACM0002 has been used which is ver 20.0.	OK	OK
If there are deviations from the methodology, has the evidence been provided on the following?	VCS Std. Version 4.0	DR	N/A	OK	OK
3.6.2.1. The deviation will not negatively impact the conservativeness of the quantification of GHG emission reductions or removals.	VCS Std. Version 4.0	DR	N/A	OK	OK
3.6.2.2. The deviation relates only to the criteria and procedures	VCS Std.	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
for monitoring or measurement, and does not relate to any other part of the methodology	Version 4.0				
4. **QUANTIFICATION of GHG EMISSION REDUCTIONS and REMOVALS**					
4.1. **Baseline Emissions**					
Do the steps taken and equations applied to calculate baseline emissions comply with the requirements of the selected baseline and monitoring methodology including applicable tool(s)?	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	Equations are provided in PD, they are correct	OK	Ok
Is it explained and clearly stated how the procedures in the approved methodology to calculate baseline emissions are applied by the PPs?	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	Selections explained	OK	OK
4.2. **Project Emissions**					
Do the steps taken and equations applied to calculate project emissions comply with the requirements of the selected baseline and monitoring methodology including applicable tool(s)?	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	Equations are provided in PD, they are correct	OK	Ok

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
Is it explained and clearly stated how the procedures in the approved methodology to calculate project emissions are applied by the PPs?	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	a) Capacity and area are also monitored in CP1, please include in PDD	CAR-3	OK
Has the PPs explained and justified all relevant methodological choices including the following?	CDM-PDD-FORM Version 11.0 EB101 Report Annex 1 §72	DR	N/A	OK	OK
4.2.3.1. Where the methodology(ies) or standardized baseline(s) include different scenarios or cases, indicate and justify which scenario or case applies to the project activity	CDM-PDD-FORM version 11.0 EB101 Report Annex 1 §72	DR	N/A	OK	OK
4.2.3.2. Where the methodology(ies) or standardized baseline(s) provide different options to choose from , indicate and justify which option is chosen for the project activity	CDM-PDD-FORM version 11.0 EB101 Report Annex 1	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
	§72				
4.2.3.3. Where the methodology(ies) or standardized baseline(s) allow different default values, indicate and justify which of the default values have been chosen for the project activity.	CDM-PDD-FORM Version 11.0	DR	N/A	OK	OK
4.3. **Leakage**					
Do the steps taken and equations applied to calculate leakage comply with the requirements of the selected baseline and monitoring methodology including applicable tool(s)?	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	N/A	OK	OK
Is it explained and clearly stated how the procedures in the approved methodology to calculate leakages are applied by the PPs?	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	N/A	OK	OK
4.4. **Summary of GHG Emission Reductions and Removals**					
Have the project proponents included the description of the procedure for quantification of the net GHG emission reductions and removals including all relevant	VCS Std. Version 4.0	DR	Yes	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
equations?					
Are the ex-ante calculation (estimate) of baseline emissions/removals, project emissions/removals, leakage emissions and net emission reductions and removals provided in a tabular format?	VCS Std. Version 4.0	DR	Yes	OK	OK
Has it been documented how each equation is applied in a manner that enables the reader to reproduce the calculation?	VCS Std. Version 4.0	DR	Yes	OK	OK
Are the example calculations for all key equations provided to allow the reader to reproduce the calculation of estimated net GHG emission reductions or removals?	VCS Std. Version 4.0	DR	Yes	OK	OK
ACM 0002					
Are baseline emissions calculated using equation (11) given in the methodology?	ACM 0002 Version 20.0	DR	Yes	OK	OK
Is the quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y ($EG_{PJ,y}$) calculated using equations (12), (13), (14), (15) or (16) given in the methodology depending on the project type and relevant requirements?	ACM 0002 Version 20.0	DR	Yes	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
• • ➢ ➢ ➢					
When the methodology offers options for approaches in calculations, is it documented in the PD which option is applied?	ACM 0002 Version 17.0	DR	Yes	OK	OK
In the case of retrofits or replacements, has the point in time when the existing equipment would need to be replaced/retrofitted in the absence of the project chosen in a conservative manner?	ACM 0002 Version 20.0	DR	N/A	OK	OK
In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects)	ACM 0002 Version 20.0	DR	N/A	OK	OK
4.4.9.1. Is it ensured that the existing plant started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions?	ACM 0002 Version 20.0	DR	N/A	OK	OK
4.4.9.2. Is it defined in the baseline emission section that no capacity addition, retrofit or rehabilitation of the plant has	ACM 0002 Version	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
been undertaken between the start of this minimum historical reference period and the implementation of the project activity?	20.0				
Are the project emissions calculated properly using equations (1), (2), (3), (4), (5), (6), (7), (8), (9) or (10) given in the methodology depending on the project type and the power density value?	ACM 0002 Version 20.0	DR	Yes	OK	OK
Where project emissions are taken as "0", has the PP made proper justification?	ACM 0002 Version 20.0	DR	Yes	OK	OK
Are the emission reductions calculated using equation (17) given in the methodology?	ACM 0002 Version 20.0	DR	Yes	OK	OK
ACM 0001					
Are the baseline emissions calculated using relevant equations from equation (1) to equation (21) in the methodology? • • • •	ACM 0001 Version 19.0	DR	N/A	OK	OK
Are the project emissions calculated using relevant equations from equation (22) to equation (25) in the	ACM 0001 Version	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
methodology?	19.0				
Are the emission reductions calculated using equation (26) in the methodology?	ACM 0001 Version 19.0	DR	N/A	OK	OK
AM0058					
Are the baseline emissions calculated using equation (1) in the methodology?	AM0058 Version 5.0	DR	N/A	OK	OK
Are the baseline emissions from heat generation calculated using equation (2) in the methodology?	AM0058 Version 5.0	DR	N/A	OK	OK
Is the CO2 emission factor for heat supply in the baseline calculated using equation (3) in the methodology?	AM0058 Version 5.0	DR	N/A	OK	OK
Is the emission factor for new users calculated using equation (4) in the methodology?	AM0058 Version 5.0	DR	N/A	OK	OK
Are the baseline emissions from the power generation calculated using equation (5) and equation (6) in the methodology?	AM0058 Version 5.0	DR	N/A	OK	OK
Are the project emissions calculated using latest applicable version of "Tool To Calculate Project or Leakage CO2 Emissions From Fossil Fuel Combustion" and the relevant principles defined in the	AM0058 Version 5.0	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
methodology? • •					
Are the leakage emissions calculated using equation (7) and equation (8) in the methodology?	AM0058 Version 5.0	DR	N/A	OK	OK
Are the emission reductions calculated using equation (9) in the methodology?	AM0058 Version 5.0	DR	N/A	OK	OK
AMS I.C.					
Are the baseline emissions for electricity produced in captive plants calculated using equation (1) given in the methodology?	AMS I.C. Version 21 §32	DR	N/A	OK	OK
Are the baseline emissions for supply of electricity to and/or displacement of electricity from a grid calculated using equation (2) given in the methodology?	AMS I.C. Version 21 §33	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
Are the baseline emissions for thermal energy produced using fossil fuels and/or grid electricity calculated using equation (3) given in the methodology?	AMS I.C. Version 21 §34 §35	DR	N/A	OK	OK
Is the emission factor of electricity determined according to the following?	AMS I.C. Version 21 §36	DR	N/A	OK	OK
4.4.27.1. For project activities that displace on-site captive electricity and/or displace grid electricity import and/or supply electricity to a grid, does the electricity emission factor reflect the emissions intensity of the captive power plant and the grid of the baseline scenario?	AMS I.C. Version 21 §36	DR	N/A	OK	OK
4.4.27.2. For project activities that do not displace captive electricity generated by an existing plant but displace grid electricity import and/or supply electricity to a grid, does the emission factor of the grid calculated as per the procedures detailed in AMS-I.D or AMS-I.F?	AMS I.C. Version 21 §37	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
4.4.27.3. For new facilities, has the emission factor of electricity taken as the most conservative (lowest) emission factor of the two power sources (captive power plant and grid)?	AMS I.C. Version 21 §38	DR	N/A	OK	OK
Are the baseline emissions for electricity and thermal energy produced in a baseline cogeneration or trigeneration unit using fossil fuel calculated using equation (4) given in the methodology?	AMS I.C. Version 21 §39	DR	N/A	OK	OK
In the case of an existing baseline cogeneration or trigeneration plant, is the efficiency of the baseline units determined by adopting one of the following criteria with the exclusion of cogeneration plants?	AMS I.C. Version 21 §40	DR	N/A	OK	OK
4.4.29.1. Most recent three years operational data immediately prior to the start date of the project activity in case of existing facilities which has more than three years of operation history;	AMS I.C. Version 21 §26	DR	N/A	OK	OK
4.4.29.2. A minimum of most recent one year data in case of existing facilities which has more than three years of operation history but do not have three years operational data; or	AMS I.C. Version 21 §26	DR	N/A	OK	OK
4.4.29.3. A performance test/measurement campaign	AMS I.C.	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
carried out prior to the implementation of the project activity in case of existing facilities which has more than three years of operation history but do not have operational data/information such as efficiency, energy consumption and output or the available data is not reliable due to various factors such as the use of imprecise or non-calibrated measuring equipment. The project proponent may follow the relevant provisions from the "Tool to determine baseline efficiency of thermal and electricity systems".	Version 21 §26 §32				
4.4.29.4. For household or commercial applications/systems, whose maximum output capacity is less than 45 kW thermal and where it can be demonstrated that the metering of thermal energy output is not plausible, as in the case of cooking stoves, gasifiers, driers, water heaters etc., is the efficiency of the baseline units determined by adopting one of the following criteria?	AMS I.C. Version 21 §43	DR	N/A	OK	OK
4.4.29.5. Highest measured operational efficiency over the full range of operating	AMS I.C. Version 21	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
conditions of a representative sample of units with similar specifications, using baseline fuel	§43-a				
4.4.29.6. Highest of the efficiency values provided by two or more manufacturers for units with similar specifications using the baseline fuel	AMS I.C. Version 21 §43-b	DR	N/A	OK	OK
4.4.29.7. Highest efficiency from referenced literature values or default efficiency of 100%.	AMS I.C. Version 21 §43-c	DR	N/A	OK	OK
For the cases where electricity is imported from a grid and/or produced in an on-site captive power plant using fossil fuel thermal energy is produced using biomass, are the baseline emissions from production of electricity calculated in line with the paragraphs 29 and 30 of the methodology?	AMS I.C. Version 21 §44	DR	N/A	OK	OK
For the cases where the electricity is produced in an on-site captive power plant using biomass (with a possibility of export to a grid) and/or imported from a grid; thermal energy is produced using fossil fuel, are the baseline emissions from the production of thermal energy using fossil fuel calculated using equation (1) given in the methodology?	AMS I.C. Version 21 §32 §45	DR	N/A	OK	OK

*DR= Document Review, I= Interview, SV= Site Visit

Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
For the cases where electricity and thermal energy are produced in a biomass fired cogeneration unit and for the cases where electricity is imported from a grid and/or produced in a biomass fired cogeneration unit; steam/heat is produced in a biomass fired cogeneration unit and/or a biomass fired boiler, are the baseline emissions from the additional production of electricity that displaces grid electricity import and/or supply electricity to the grid, calculated using as per the procedures detailed in AMS-I.D or AMS-I.F as applicable?	AMS I.C. Version 21 §30 §46	DR	N/A	OK	OK
For the cases where the electricity and/or thermal energy produced in a co-fired system and other project activities where the baseline is a co-fired system, are the baseline emissions determined based on three years average historical data on the relative share of fossil fuel and biomass in the baseline fuel mix using equation (5) given in the methodology?	AMS I.C. Version 21 §47	DR	N/A	OK	OK
For the cases where the electricity is imported from a grid and/or produced in an on-site captive power	AMS I.C. Version 21	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
plant using fossil fuel and thermal energy is produced using electricity, are the baseline emissions calculated using equation (6) given in the methodology?	§48				
For the cases where the electricity is imported from a grid and/or produced in an on-site captive power plant using fossil fuel and thermal energy is produced using electricity, are the baseline electricity related emissions (BEgrid,y) calculated using equation (2) given in the methodology?	AMS I.C. Version 21 §49	DR	N/A	OK	OK
For the cases where the electricity is imported from a grid and/or produced in an on-site captive power plant using fossil fuel and thermal energy is produced using electricity, are the baseline emissions (BEcaptelec,y) from electricity obtained from captive power plant(s), calculated using equation (1) given in the methodology?	AMS I.C. Version 21 §50	DR	N/A	OK	OK
For the cases where the electricity is imported from a grid and/or produced in an on-site captive power plant using fossil fuel and thermal energy is produced using electricity, are the baseline emissions associated with the electricity consumed, whether it is from captive power plants and/or power from the grid, to produce chilled water within the project boundary are determined	AMS I.C. Version 21 §51	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
using equation (7) given in the methodology?					
For the cases where the electricity is imported from a grid and/or produced in an on-site captive power plant using fossil fuel and thermal energy is produced using electricity, is the baseline scenario chiller COP (COPc,i) determined in the following manner?	AMS I.C. Version 21 §51-a	DR	N/A	OK	OK
4.4.38.1. If the baseline scenario is an existing chiller or chillers, then the COP shall be based on existing chiller performance data as specified in paragraph 25 of the methodology	AMS I.C. Version 20 §51-a-i	DR	N/A	OK	OK
4.4.38.2. If the baseline scenario is a chiller or chillers that would have been built (i.e. not existing chillers), the COP shall be determined as the highest COP full load performance value provided by two or more manufacturers for chillers commonly sold in the project country for the indicated commercial application;	AMS I.C. Version 21 §51-a-ii	DR	N/A	OK	OK
4.4.38.3. For the cases where the electricity is imported from a grid and/or produced in an on-site captive power plant using fossil fuel and thermal energy is produced using electricity, is the cooling output of each	AMS I.C. Version 21 §51-b	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
baseline scenario chiller calculated using equation (8) given in the methodology?					
For project activities with water heating systems, that use electricity, are the baseline emissions determined using equation (9) given in the methodology?	AMS I.C. Version 21 §52	DR	N/A	OK	OK
For the cases where project activities involve the addition of renewable energy units at an existing renewable energy production facility, are the existing units considered in the determination of baseline emissions, project emissions, and/or leakage, as relevant?	AMS I.C. Version 21 §53	DR	N/A	OK	OK
In the cases where project activities involve the addition of new energy production units at an existing facility has the net increase in thermal energy been calculated using equations (10) and (11) given in the methodology?	AMS I.C. Version 21 §54 §55 §56 §57	DR	N/A	OK	OK
If the project activity seeks to retrofit or modify an existing facility to enhance the energy conversion efficiency, is the thermal energy production by an existing facility calculated using equation (12) given	AMS I.C. Version 21 §59	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
in the methodology?					
If the project activity seeks to retrofit or modify an existing facility to enhance the energy conversion efficiency, are the baseline emissions from the incremental thermal energy supplied due to retrofit calculated using equation (13) given in the methodology?	AMS I.C. Version 21 §60	DR	N/A	OK	OK
If the project activity seeks to retrofit or modify an existing facility to enhance the energy conversion efficiency, are the requirements concerning the demonstration of the remaining lifetime of the replaced equipment met as described in the "General Guidelines for SSC CDM methodologies"?	AMS I.C. Version 21 §61 §62	DR	N/A	OK	OK
In case of the project activities that seek to retrofit or modify an existing facility for the purpose of fuel switch from fossil fuels to biomass in heat generation equipment, are the baseline emissions calculated using equation (2) given in the methodology?	AMS I.C. Version 21 §63	DR	N/A	OK	OK
In case of household or commercial applications/systems, whose maximum output capacity is less than 45 kW thermal and where it can be demonstrated that the metering of thermal energy output is not plausible, as in the case of biomass stoves, gasifiers, driers, water	AMS I.C. Version 21 §64	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
heaters etc., are the baseline emissions calculated as per equation (14) given in the methodology?					
If applicable, are the quantities and types of biomass and the biomass to fossil fuel ratio (in the case of co-fired systems) to be used during the crediting period explained and documented transparently in the CDM-PDD?	AMS I.C. Version 21 §65	DR	N/A	OK	OK
If applicable, have the ex-ante estimation of biomass and the biomass to fossil fuel ratio quantities been provided for the selection of the baseline scenario?	AMS I.C. Version 21 §65	DR	N/A	OK	OK
If applicable, is the project emissions calculated using equation (15) in the methodology?	AMS I.C. Version 21 §66	DR	N/A	OK	OK
If applicable, are the project emissions from on-site consumption of fossil fuels calculated using the latest version of the *"Tool to calculate project or leakage CO2 emissions from fossil fuel combustion"*?	AMS I.C. Version 21 §67	DR	N/A	OK	OK
If applicable, are the project emissions resulting from the electricity consumption by the project activity calculated using the latest version of the "*Tool to calculate baseline, project and/or leakage emissions from electricity consumption*"?	AMS I.C. Version 21 §68	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
In case of geothermal project activities, have the PPs accounted for the following emission sources, where applicable?	AMS I.C. Version 21 §69	DR	N/A	OK	OK
4.4.52.1. Fugitive emissions of carbon dioxide and methane due to release of non-condensable gases from produced steam	AMS I.C. Version 21 §69	DR	N/A	OK	OK
4.4.52.2. Carbon dioxide emissions resulting from combustion of fossil fuels related to the operation of the geothermal power plant	AMS I.C. Version 21 §69	DR	N/A	OK	OK
In case of geothermal project activities, are the project emissions calculated using equations (16), (17) and (18) given in the methodology?	AMS I.C. Version 21 §70 §71 §72	DR	N/A	OK	OK
In case of trigeneration project activities, are the project emissions calculated using equations (19) and (20) and the options given in the methodology?	AMS I.C. Version 21 §73 §74 §75	DR	N/A	OK	OK
In cases where the project activity utilizes biomass sourced from dedicated plantations, are the project emissions from biomass cultivation calculated according to the latest version of “Project emissions from cultivation of biomass” tool?	AMS I.C. Version 21 §76	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
In cases where the energy generating equipment currently being utilised is transferred from outside the boundary to the project activity, is the leakage considered?	AMS I.C. Version 21 §77 §80	DR	N/A	OK	OK
In cases where the collection /processing /transportation of biomass residues is outside the project boundary and due to the implementation of the project activity biomass residues are transported over a distance of 200 kilometres, are the CO2 emissions from the collection, processing and transportation of biomass residues to the project site calculated using the latest version of tool "Project and leakage emissions from transportation of freight"?	AMS I.C. Version 21 §78	DR	N/A	OK	OK
In case of biomass project activities, are the leakage emissions on account of the diversion of biomass residues from other uses calculated in line with the latest version of "General guidance on leakage in biomass project activities"?	AMS I.C. Version 21 §80	DR	N/A	OK	OK
Have the PPs calculated the emission reductions using equation (21) given in the methodology?	AMS I.C. Version 21 §81	DR	N/A	OK	OK
AMS I.D.					

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
Are baseline emissions calculated using equation (1) given in the methodology?	AMS I.D. Version 18.0 §22	DR	N/A	OK	OK
Is the emission factor calculated using one of the following options:	AMS I.D. Version 18.0 §23	DR	N/A	OK	OK
4.4.61.1. A combined margin (CM), consisting of the combination of operating margin (OM) and build margin (BM) according to the procedures prescribed in the “Tool to calculate the Emission Factor for an electricity system	AMS I.D. Version 18.0 §23	DR	N/A	OK	OK
4.4.61.2. The weighted average emissions (in t CO2/MWh) of the current generation mix.	AMS I.D. Version 18.0 §23	DR	N/A	OK	OK
Have the calculations been based on data from an official source (where available) and made publicly available?	AMS I.D. Version 18.0 §24	DR	N/A	OK	OK
In case of green field power plant, is the generated electricity as a result of project activity calculated using equation (2) given in the methodology?	AMS I.D. Version 18.0 §26	DR	N/A	OK	OK
In case of capacity addition in wind, solar, wave or tidal power plants, are the baseline emissions calculated using equation (3) given in the methodology?	AMS I.D. Version 18.0 §27	DR	N/A	OK	OK
In case of capacity addition in hydro or geothermal power plants, have	AMS I.D. Version	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
the requirements defined in Section 5.5.1.3 of the methodology been followed?	18.0 §28				
In case of capacity addition to biomass power plants, are the baseline emissions calculated using equations (4) and (5) given in the methodology?	AMS I.D. Version 18.0 §29 §30	DR	N/A	OK	OK
In case of retrofit, rehabilitation or replacement in hydro, solar, wind, geothermal, wave and tidal plants, are the baseline emissions calculated using equation (6) given in the methodology?	AMS I.D. Version 18.0 §31	DR	N/A	OK	OK
In case of retrofit, rehabilitation or replacement in biomass plants, are the baseline emissions calculated using equations (7) and (8) given in the methodology?	AMS I.D. Version 18.0 §32	DR	N/A	OK	OK
In case of retrofit, rehabilitation or replacement, have the PPs used among the following two time spans of historical data to determine EGhistorical?	AMS I.D. Version 18.0 §33 §35 §36	DR	N/A	OK	OK
4.4.69.1. The three last calendar years (five calendar years for hydro project) prior to the implementation of the project activity	AMS I.D. Version 18.0 §35	DR	N/A	OK	OK
4.4.69.2. The time period from the	AMS I.D.	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
calendar year following *DATEhist*, up to the last calendar year prior to the implementation of the project, as long as this time span includes at least three calendar years (five calendar years for hydro project), where *DATEhist* is latest point in time between:	Version 18.0 §35				
4.4.69.2.1. The commercial commissioning of the plant/unit;	AMS I.D. Version 18.0 §35	DR	N/A	OK	OK
4.4.69.2.2. If applicable: the last capacity addition to the plant/unit; or	AMS I.D. Version 18.0 §35	DR	N/A	OK	OK
4.4.69.2.3. If applicable: the last retrofit of the plant/unit	AMS I.D. Version 18.0 §35	DR	N/A	OK	OK
In case of retrofit, rehabilitation or replacement, have PPs followed the latest applicable version of "Tool to determine the remaining lifetime of equipment" to estimate DATEBaselineRetrofit? DATEBaselineRetrofit is the point in time when the existing equipment would need to be replaced/retrofitted in the absence of the project activity. The point in time when the existing equipment would need to be replaced/retrofitted in the absence of the project activity should be chosen in a conservative manner that is, if a range is	AMS I.D. Version 18.0 §37 §38	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
identified, the earliest date should be chosen.					
Where the project emissions are taken as "0" have the PPs made proper justification?	AMS I.D. Version 18.0 §39	DR	N/A	OK	OK
If the proposed project activity is a geothermal power plant or a hydropower plant, have the project emissions been considered following the procedure described in most recent version of ACM0002?	AMS I.D. Version 18.0 §39	DR	N/A	OK	OK
If necessary, have the PPs calculated the CO2 emissions from on-site consumption of fossil fuels due to the project activity using the latest applicable version of the "Tool to calculate project or leakage CO2 emissions from fossil fuel combustion?	AMS I.D. Version 18.0 §40	DR	N/A	OK	OK
In case biomass is sourced from dedicated plantations, have the procedures in the tool "Project emissions from cultivation of biomass" been followed to calculate project emissions?	AMS I.D. Version 18.0 §41	DR	N/A	OK	OK
Has the general guidance on leakage in biomass project activities been followed to quantify leakages pertaining to the use of biomass residues?	AMS I.D. Version 18.0 §42	DR	N/A	OK	OK
Are the emission reductions calculated using equation (9) given in the methodology?	AMS I.D. Version 18.0 §43	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
5. **MONITORING**					
5.1. **Data and Parameters That Are Available At Validation**					
Are all data sources and assumptions for the ex-ante values, appropriate and correct?	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	Please update 5.1 CM	CAR-4	OK
Are all ex-ante values applicable to the proposed VCS project activity?	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	Yes	OK	OK
Are all ex-ante values resulting in an accurate or otherwise conservative estimate of the emission reductions?	CDM-PDD-FORM Version 11.0	DR	Yes	OK	OK
Are the ex-ante data and parameters correct?	CDM-PDD-FORM Version 11.0	DR	Yes	OK	OK
Are the units of the data and parameters specified correctly?	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	Yes	OK	OK
Are the description of data and parameters given correctly?	VCS Std. Version 4.0	DR	Yes	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
	CDM-PDD-FORM Version 11.0				
Is the choice of the source of data explained and justified?	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	Yes	OK	OK
Are the applied actual values provided correctly?	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	Yes	OK	OK
Is the choice of data explained and justified?	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	Yes	OK	OK
Where values have been measured, has a description of the measurement methods and procedures (e.g. which standards have been used) been included?	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	N/A	OK	OK
Where values have been measured, are the following included in the VCS PD?	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
5.1.11.1. The equipment used	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	N/A	OK	OK
5.1.11.2. responsible person/entity having undertaken the measurement	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	N/A	OK	OK
5.1.11.3. the date of measurement(s)	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	N/A	OK	OK
5.1.11.4. the frequency of measurement(s)	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	N/A	OK	OK
5.1.11.5. the measurement results?	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	N/A	OK	OK
5.2. **Data and Parameters Monitored**					

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
In the data/parameter tabular formats for monitoring, has the name of the each data/parameter been included?	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0 AM0058 Version 5.0	DR	Please refer below	OK	OK
Has the unit of the each data/parameter been included?	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	Included	OK	OK
Has the description of the each data/parameter been included?	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	Included	OK	OK
Has the source of the each data/parameter been included?	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	Included	OK	OK
Where several sources of data/parameters are used, is the choice of data sources explained and justified?	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	N/A	OK	OK
Has the frequency of	VCS Std.	DR	Continuous measurement / Monthly recording	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
monitoring/recording been included?	Version 4.0				
Are the applied actual values provided correctly?	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	provided correctly	OK	OK
Has the measurement methods and procedures been included?	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	Included	OK	OK
Has the PPs included which measurement equipment is used for monitoring?	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	a) Included	OK	OK
Has the PPs included how the measurement is undertaken?	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	Included	OK	OK
Have the PPs included description of calibration procedures for the monitoring equipment?	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	Included	OK	OK
Has the accuracy level of the measurement method included?	VCS Std. Version 4.0 CDM-PDD-	DR	Please refer above	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
	FORM Version 11.0				
Has the responsible person/entity and the interval for the measurements included?	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	Please refer above	OK	OK
If applicable, has the calculation method been included?	VCS Std. Version 4.0	DR	N/A	OK	OK
If the data and parameters monitored in Section 5.2 of the PD are to be determined by a sampling approach, has the PP provided a description of the sampling plan in accordance with the recommended outline for a sampling plan in the latest applicable version of "Standard for Sampling and Surveys for CDM Project Activities and Programme of Activities"? • • • • (	CDM-PDD-FORM version 11.0 EB105 Report Annex 1 §29 §30 §31 §32 §33	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
• • •					
If the sampling approach is used by the PPs, does the sampling plan present a reasonable approach for obtaining unbiased, reliable estimates of the variables?	EB86 Report Annex 4 §40a	DR	N/A	OK	OK
If the sampling approach is used by the PPs, are the elements of objectives and reliability requirements complete?	EB86 Report Annex 4 §40a-i	DR	N/A	OK	OK
If the sampling approach is used by the PPs, do the requirements specified agree with those stated in the appropriate standards?	EB86 Report Annex 4 §40a-i	DR	N/A	OK	OK
If the sampling approach is used by the PPs, is the population in the sampling plan clearly defined?	EB86 Report Annex 4 §40b	DR	N/A	OK	OK
If the sampling approach is used by the PPs, is the proposed sampling approach clear?	EB86 Report Annex 4 §40c	DR	N/A	OK	OK
If the sampling approach is used by the PPs, does the sampling approach comply with the description of the population?	EB86 Report Annex 4 §40c-ii	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
If the sampling approach is used by the PPs, is the proposed sample size adequate to achieve the minimum confidence/precision requirements?	EB86 Report Annex 4 §40d	DR	N/A	OK	OK
If the sampling approach is used by the PPs, is the ex-ante estimate of the population variance needed for the calculation of the sample size adequately justified?	EB86 Report Annex 4 §40d	DR	N/A	OK	OK
If the sampling approach is used by the PPs, is the sample representative of the population?	EB86 Report Annex 4 §40e	DR	N/A	OK	OK
If the sampling approach is used by the PPs, is it identified how the sampling frame would be kept?	EB86 Report Annex 4 §40e-ii	DR	N/A	OK	OK
If the sampling approach is used by the PPs, are the methods of data collection clear and unambiguous?	EB86 Report Annex 4 §40f-i	DR	N/A	OK	OK
If the sampling approach is used by the PPs, are the procedures for the data measurements defined appropriately and clearly?	EB86 Report Annex 4 §40g	DR	N/A	OK	OK
If the sampling approach is used by the PPs, do the procedures for measurements adequately provide	EB86 Report Annex 4	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
for minimizing non-sampling errors?	§40g				
If the sampling approach is used by the PPs, is the quality control and assurance strategy adequate?	EB86 Report Annex 4 §40g-i	DR	N/A	OK	OK
If the sampling approach is used by the PPs, are the proposed skill sets, qualifications and experience of the personnel to be engaged to conduct sampling adequate?	EB86 Report Annex 4 §40h-i	DR	N/A	OK	OK
5.3. **Monitoring Plan**					
Are the organizational structure, responsibilities and competencies of the personnel that will be carrying out monitoring activities identified?	VCS Std. Version 4.0	DR	Responsibilities explained	OK	OK
Are the methods for generating, recording, storing, aggregating, collating and reporting data on monitored parameters described?	VCS Std. Version 4.0	DR	Responsibilities explained	OK	OK
Are the policies for oversight and accountability of monitoring activities described?	VCS Std. Version 4.0	DR	N/A	OK	OK
Are the procedures for handling internal auditing and non-conformities described?	VCS Std. Version 4.0	DR	N/A	OK	OK
If the sampling approaches used in the monitoring plan, has the following been included?	VCS Std. Version 4.0	DR	N/A	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
5.3.5.1. target precision levels	VCS Std. Version 4.0	DR	N/A	OK	OK
5.3.5.2. sample sizes	VCS Std. Version 4.0	DR	N/A	OK	OK
5.3.5.3. sample site locations	VCS Std. Version 4.0	DR	N/A	OK	OK
5.3.5.4. stratification	VCS Std. Version 4.0	DR	N/A	OK	OK
5.3.5.5. frequency of measurement and	VCS Std. Version 4.0	DR	N/A	OK	OK
5.3.5.6. QA/QC procedures.	VCS Std. Version 4.0	DR	N/A	OK	OK
Have the PPs developed and described the monitoring plan for the proposed project activity in accordance with the selected methodology(ies) and all other applicable rules and requirements?	EB93 Report Annex 4 §75 EB93 Report Annex 5 §117	DR	Yes	OK	OK
Does the monitoring plan include all data, parameters and related information required by the selected methodology(ies)?	EB93 Report Annex 4 §76 EB93 Report	DR	Yes	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
	Annex 5 §118a-ii ACM 0002 Version 17.0				
Are the monitoring arrangements described in the monitoring plan feasible within the project design?	EB93 Report Annex 5 §118b	DR	Yes	OK	OK
Is the operational and management structure for the monitoring of emission reductions or any leakage emissions described in the monitoring plan?	CDM-PDD-FORM Version 11.0	DR	Yes	OK	OK
Does the monitoring plan clearly indicate the responsibilities and internal arrangements for data collection and archiving?	CDM-PDD-FORM Version 11.0 EB101 Report Annex 1 §82c	DR	Yes	OK	OK
Does the monitoring plan reflect good monitoring practices appropriate to the type of project activity?	CDM-PDD-FORM Version 11.0	DR	Yes	OK	OK
Are measurements conducted with according to relevant industry or national/international standards?	CDM-PDD-FORM Version 11.0	DR	Yes	OK	OK
Where appropriate, are the line diagrams to display the GHG data collection and management system	VCS Std. Version 4.0	DR	Yes	OK	OK

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Question	Reference	Means of validation*	Findings, comments, references and document sources	Draft opinion	Final opinion
been included?					
Annex-1 Baseline information					
1. If any further background information regarding the application of the baseline methodology is provided, is this information correct and supported by the appropriate evidence?	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	N/A	OK	OK
Annex-2 Monitoring information					
1. If any further background information regarding the application of the monitoring methodology is provided, is this information correct and supported by the appropriate evidence?	VCS Std. Version 4.0 CDM-PDD-FORM Version 11.0	DR	N/A	OK	OK

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Table 2 – Resolution of Corrective Action, Forward Action and Clarification Requests

Draft Report Clarifications, Forward Action and Corrective Action Requests by Validation Team	Ref. to Checklist Questions in Table-1	Summary of Project Participants' Response	Validation Team Conclusion
CAR-1 Please use appropriate coefficients for OM and BM and revise CM and all relevant numbers in PDD	1.1.1.5	CM and the values have been revised accordingly.	Review 1: Ok closed (Corrected)
CAR-2 Please revise Erdemli transformer	3.3	Revised accordingly	Review 1: Ok closed (Corrected)
CAR-3 Capacity and area are also monitored in CP1, please include in PDD	4.2	Since the power density is still greater than 10W/m2, CAPPJ and APJ have not been put as monitored parameters during the second crediting period.	Review 1: Ok closed (Explained)
CAR-4 Please update 5.1 CM	5.1	Updated.	Review 1: Ok closed (Corrected)
CL-1 a-Please include timeline of project b-Please state turbine capacities in Mwe and MWm c-Capacities do not match with license, please explain	1.1	a- It's been added. b-It's been revised accordingly. c- The type has been revised accordingly.	Review 1: a)Ok closed (Added) b) Ok closed (Stated) c) Ok closed (Corrected)
CL-2 Please explain the below: -procedures or methods used for engaging local stakeholders currently -mechanism for on-going communication -how comments are taken into account	2.2	Section 2.2. has been revised accordingly.	Review 1: Ok closed (Added)
CL-3	2.3	It's been provided.	Review 1:

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Please provide EIA not required certificate			Ok closed (Provided)
CL-4 No environmental impacts stated a-Please provide explanation in PD 2.3 about waste management activities on site, minimum water applications, fish passage, trainings, local employment. Please also mention the dates and details of evidences sent to DOE b- For CTP pipes please state glass fiber reinforced polyester pipes c-Sentence not clear: The only project item whose excavation activities has not started is Uzundere-I weir.	2.3	Section 2.3 has been revised accordingly.	Review 1: a-b-c)Ok closed (Corrected)
CL-5 Please calculate and show power density in excel and PDD	3.2.4.3	It's been added.	Review 1: Ok closed (Shown)
CL-6 Please confirm in the Section 3.5 of the PD that project is not being implemented to comply with any law or other regulatory framework.	3.5	Section 3.5 has been revised accordingly.	Review 1: Ok closed (Stated)
CL-7 PDD (Section 1.1): As per the Section 1.1 of the latest VCS-PD (2nd CP), "total installed capacity in the powerhouse is 63.068MWm / 62.152 MWe". As per the Section 1.11 of VCS PD, "the Uzundere I 63.0 MW Hydroelectric Power Project entails the construction and operation of a 63 MW hydroelectric power plant, the construction of a transmission line, connection canals, concrete tunnels and penstock". While as per the previous		The project has not been changed from 63.0 MW to 63.068MWm. Probably writing difference has been realized.	Review 1: Ok closed (Explained)

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registered VCS PD (1st CP), "In the power house, two vertical shaft Pelton turbines will be installed, each with 31.6 MW", which means total capacity of 63.2 MW. Check on the differences observed and also clarify whether any changes happened in the project capacity during this 2nd CP as compared to 1st CP. Clarify.			

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CL-8 Both the version ('version 7.0.0' and 'version 5.2') of "Tool for the demonstration of and assessment of additionality" has been used in the Section 3.5 & 3.1 of the PDD. Check.		Since version 5.2. has been used during the investment analysis of the project activity, this version number was kept. However, it's been omitted.	Review 1: Ok closed (Stated)
CL-9 MR (Section 3.2): For the "Applicability Criteria No. 3", the "Justification" has been mentioned as "The project is not a hydro power plant that results in a new reservoir or in the increase in existing reservoirs where the power density of the power plant is less than 4 W/m2". However as per the requirements of this "Applicability Criteria", the condition of "power density of the power plant less than 4 W/m2" is applicable when "the project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs". Check.		Revised accordingly.	Review 1: Ok closed (Corrected)
CL-10 The heading of this Section (Project Boundary) has been disturbed. The original heading format shall be restored in the PDD:		It's in the same format.	Review 1: Ok closed (Corrected)
CL-11 As per the requirements of the applied methodology, the following parameters are to be determined "Once at the beginning of each crediting period". - CapPJ (Installed capacity of the hydro power plant after the implementation of the project activity)		Explanation has been added 5.3.	Review 1: Ok closed (Parameters omitted)

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- APJ (Area of the single or multiple reservoirs measured in the surface of the water, after the implementation of the project activity, when the reservoir is full) Hence the value of each of the above parameters shall be clearly indicated in then VCS-PD.			

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CL-12 ERs Excelsheet: Please refer to the “Cell C8” in the “Combined Margin EF” spreadsheet of the ERs Excelsheet, where the unit is presented as “tCO3”, which shall be corrected.		Corrected.	Review 1: Ok closed (Corrected)
CL-13 ERs Excelsheet: Please refer to the “Cell D16” in the “Combined Margin EF” spreadsheet of the ERs Excelsheet, where the total emission reductions formula is available as: =ROUNDDOWN(SUM(E16:G16),0) However as per the applied methodology, the Emission Reductions = Baseline Emissions – Project Emissions – Leakage Emissions Hence appropriate corrections shall be provided in the ERs Excelsheet.		Corrected. However, there was no difference, so it was left.	Review 1: Ok closed (Corrected)
CL-14 Please revise site visit date in PDD (not 2021)		Corrected.	Review 1: Ok closed (Corrected)
CL-15 The value (15,65%) shall be corrected in the statement “the IRR of the project was compared against an investment benchmark which is derived by the CAPM scientific model and was calculated as 15,65 %”. The correct decimal (.) and comma (,) format shall be used in the PDD		Corrected.	Review 1: Ok closed (Corrected)

* CAR= Corrective Action Request, FAR= Forward Action Request, CL= Clarification Request

ANNEX 2: VALIDATION TEAM AND ITR COMPETENCE

Mrs. Fikriye Seda ATABEK holds B.Sc. degree in “Chemical Engineering” and a M.Sc. degree in “Energy Science and Technology”. She is a lead auditor and trainer for ISO 50001 and since 2004 has been working in the fields of “Management systems”, “ISO 14064” and “Energy Management in Industry”. She has been involved in many GS and VCS projects as ITR, Team Leader, Validator or Verifier since 2010. With re-carbon, Seda is a free-lance Team Leader and ITR.

Ms. Öykü YAKUPOĞLU holds a B.Sc. degree in “Environmental Engineering” from Middle East Technical University/Ankara and currently undergoes a M.Sc. program in Chemical Engineering. She is experienced in ISO 14001: 2015 - Environment Management System, ISO 50001: 2018- Energy Management System , ISO 45001: 2018 - Occupational Health and Safety, Management System, ISO 9001: 2015 - Quality Management System Internal Auditor, ISO 14001: 2015 - Environment Management System Internal, Auditor, ISO 50001: 2018- Energy Management System Internal Auditor. With re-carbon, Öykü is a Validator/Verifier Trainee.

Mr. Rohit BADAYA holds Masters degree (M. Tech) in Nanotechnology and a Bachelors degree (B.Tech) in Pulp and Paper Engineering from the Indian Institute of Technology Roorkee (IITRoorkee). He is also an Energy Auditor, cerified by the Bureau of Energy Efficiency, Ministry of Power, Govt. of India. Rohit has more than 13 years of work experience in the area of Climate Change (CDM, GS, VCS) and has worked for various DOEs/VVBs in the past, including “TÜV Nord”, “PJRCES Inc.” and “KBS Certification Services Private Limited”, where he worked as a Team Leader, Validator/Verifier, Technical Expert, ITR, Manager (Technical & Certification) and Quality Manager. Within the context of CDM/GS/VCS, Rohit is a Technical Expert for Technical Areas TA 1.1 (Thermal energy generationfrom fossil fuels and biomass including thermal electricity from solar), TA 1.2 (Energy generation from renewable energy sources), TA 2.1 (Energy Distribution), TA 3.1 (Energy Demand), TA 13.1 (Waste Handling and Disposal), TA 13.2 (Manure). Rohit has a track record of more than 200 projects as Team Leader, Validator, Verifier, Technical Expert and Technical Reviewer. He is well versed with various local regulations related to CDM/GS/VCS projects, located in countries in Africa, Asia as well as in Turkey. With re-carbon, Rohit is a free-lance Team Leader and ITR.

Annex 2-1: Appointment Certificates

Re Carbon Gözetim Denetim ve Belgelendirme Ltd. Şti. Prof. Dr. Aziz Sancar Cad. 27/6 TR / 06690 Çankaya-Ankara Tel.: 0090-312-287 5122 Fax: 0090-312-287 3373	**Certificate of Appointment** Carbon Division	re-carbon Page: 1/1

This Certificate of Appointment is given to **Mrs. Fikriye Seda ATABEK** as a confirmation of compliance with internal qualification requirements as follows:

Clean Development Mechanism				
Validator	**Verifier**	**Team leader**	**Technical reviewer**	**Technical Expert**
08-02-2021	08-02-2021	N/A	N/A	05-05-2017

Verified Carbon Standard, Gold Standard, World Commission on Dams, Social Carbon				
Validator	**Verifier**	**Team leader**	**Technical reviewer**	**Technical Expert**
08-02-2021	08-02-2021	08-02-2021	08-02-2021	08-02-2021

Speciality	**Regional (Country) expertise**	**Financial expertise**	**Technical area**
N/A	Turkey	N/A	1.2, 2.1 and 3.1

Within the scope and in strict accordance to the appointment indicated above, the bearer can:

1. Participate in the assessments conducted by Re Carbon Ltd.
2. Take the roles within and outside of the assessment team
3. Bring specific expertise to the assessments

This Certificate of Appointment is valid unless there are changes in the related requirements for the qualification and appointment and/or the personnel's work agreement is terminated and there is no defined validity period for this Certificate.

However, The Certificate may be updated, suspended or cancelled at any time, as a result of the performance assessments and/or other reasons as defined above.

APPOINTMENT IS GRANTED BY			
Mr. Anıl SÖYLER	Certification Manager	08-02-2021	
Name	**Position**	**Date**	**Signature**

[illegible]

Re Carbon Gözetim Denetim ve Belgelendirme Ltd. Şti. Prof. Dr. Aziz Sancar Cad. 27/6 TR / 06690 Çankaya-Ankara Tel.: 0090-312-287 5122 Fax: 0090-312-287 3373	**Certificate of Appointment**	re-carbon
	Carbon Division	Page: 1/1

This Certificate of Appointment is given to **Mr. Rohit BADAYA** as a confirmation of compliance with internal qualification requirements as follows:

Clean Development Mechanism				
Validator	**Verifier**	**Team leader**	**Technical reviewer**	**Technical Expert**
25-10-2021	25-10-2021	25-10-2021	25-10-2021	25-10-2021

Verified Carbon Standard, Gold Standard, World Commission on Dams, Social Carbon				
Validator	**Verifier**	**Team leader**	**Technical reviewer**	**Technical Expert**
25-10-2021	25-10-2021	25-10-2021	25-10-2021	25-10-2021

Speciality	**Regional (Country) expertise**	**Financial expertise**	**Technical area**
N/A	India and Turkey	N/A	1.1, 1.2, 2.1, 3.1, 13.1 & 13.2

Within the scope and in strict accordance to the appointment indicated above, the bearer can:

1. Participate in the assessments conducted by Re Carbon Ltd.
2. Take the roles within and outside of the assessment team
3. Bring specific expertise to the assessments

This Certificate of Appointment is valid unless there are changes in the related requirements for the qualification and appointment and/or the personnel's work agreement is terminated and there is no defined validity period for this Certificate.

However, The Certificate may be updated, suspended or cancelled at any time, as a result of the performance assessments and/or other reasons as defined above.

APPOINTMENT IS GRANTED BY			
Mr. Anıl SÖYLER	Certification Manager	25-10-2021	
Name	**Position**	**Date**	**Signature**

F-C-053 / 03.02.2021 - 02